

CSE 3105 / CSE 3135 /CSE 3137

OBJECT-ORIENTED ANALYSIS AND DESIGN

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COURSE PROJECT: *CampusSignal*

Requirements Analysis Document (RAD)

Group 1

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1 Introduction

Overview and Campus Problem

Students and staff at the Manisa Celal Bayar University campus report experiencing slow or unavailable Wi-Fi connections in classrooms, libraries, and study areas. Currently, isolated complaints fail to clearly indicate whether an issue is caused by a chronic hardware fault or temporary network congestion. Furthermore, in areas with persistently weak signals, users may stop trying to connect, resulting in a lack of complaints; consequently, IT may falsely perceive this low usage as a lack of problems.

Motivation and Objectives

The CampusSignal project aims to support the detection and resolution of Wi-Fi connectivity issues by combining on-campus connectivity measurements, access point (AP) telemetry, and a managed repair workflow. Its primary objectives are to shorten fault-detection and repair cycles, make repair ownership visible, and distinguish genuinely weak service from areas with insufficient observational data.

Success Criteria

The success of the system will be evaluated using concrete, measurable criteria based on issue history and connectivity data rather than assumptions. After the first month of operation, the median time from issue detection to technician assignment should stay below one working day, fewer than 15% of verified issues should be reopened, and at least 80% of flagged underobserved zones should reach the minimum of five fresh measurements within 14 days.

Development Context

This project is developed as part of the Object-Oriented Analysis and Design course under the "Connected Campus" theme. The scope of the project is strictly limited to the analysis and design of the software system; implementation and deployment are outside the course scope. Consequently, AP telemetry data and network measurements will be simulated to support design scenarios and UI mock-ups.

Principal Stakeholders and Actors

The primary stakeholders affected by the system include campus users (students and staff) who need reliable connectivity, university IT management who require data to optimize network resources, teaching staff whose lessons depend on network availability, and the university administration, which ensures compliance with data privacy regulations such as the Turkish Personal Data Protection Law (KVKK, Law No. 6698).

To achieve its goals, the system interacts with three distinct human actors: Campus Users (who submit evidence), IT Administrators (who review evidence and assign issues), and Technicians (who perform repairs and record interventions). Additionally, the AP Controller acts as a passive external system actor. The internal scheduler that triggers periodic evaluations is part of the system and is therefore not modelled as an actor.

System Boundary

The design covers selected campus buildings, floors, zones, and access points. The system boundary encompasses user connectivity measurements, automated issue detection, intervention recording, service recovery verification, and the targeted investigation of underobserved zones. It includes an offline queue for reports submitted without active internet and a heatmap displaying coverage quality. The system interfaces with the AP Controller strictly in a read-only capacity to retrieve online status and aggregate connected-client counts.

To protect user privacy, data collection is strictly bounded by explicit user consent, zone-level location mapping rather than precise tracking, and role-based access controls. Physical repairs of faulty devices, automatic reconfiguration of AP settings, campus-wide physical deployment, and the surveillance of users' browsing activities are strictly outside the software boundary. CampusSignal is designed as a standalone application with its own sign-in and registration. Integration into the university's existing MCBU campus mobile app is considered as a possible later deployment option and is outside the scope of this analysis.

Project Basis

The project scope and course constraints follow the OOAD Term Project Description (Fall 2026) and the team's approved CampusSignal proposal.

2 Current System

Current State: eduroam Wi-Fi Issues

Manisa Celal Bayar University provides wireless network access through eduroam, an international roaming service used by participating educational and research institutions. The university's Information Technology Department publishes connection instructions for different operating systems and provides contact information for users who require technical assistance.

Existing Workflow

Based on the university's published guidance, students experiencing Wi-Fi problems can first consult the connection instructions to resolve common configuration or authentication issues. If the problem persists, users can consult the Information Technology Department's published telephone and email contact details for assistance. The public page does not specify a dedicated ticketing workflow or guarantee that each contact becomes a trackable incident [MCBU-BID 2025].

The public guidance describes connection setup and publishes contact details, but does not document internal incident registration, assignment, prioritization, escalation, or closure. Steps after initial contact therefore remain unverified [MCBU-BID 2025].

Actors Involved

- **Students and university users:** Experience connectivity problems, attempt basic troubleshooting, and report unresolved issues.
- Information Technology Department: Publishes eduroam connection instructions and contact details. The public guidance does not describe how Wi-Fi-related contacts are logged or tracked [MCBU-BID 2025].

- Technical support personnel: May investigate reported connectivity problems, but the internal assignment and escalation process is not documented in the public material reviewed and requires confirmation.

Potential Pain Points

The use of separate communication channels, such as telephone calls and email, may create several challenges if reports are not managed through a centralized system:

- **Fragmented reporting:** Information about similar or related incidents may be distributed across different communication channels.
- **Limited status visibility:** Students may have difficulty knowing whether their reports have been received, assigned, or resolved.
- **Repeated explanations:** Users may need to describe the same problem again when communicating with different personnel.
- **Difficult prioritization:** Without a shared incident overview, identifying urgent issues or multiple reports concerning the same location may be more difficult.
- **Limited incident overview:** A centralized view of recurring issues, affected locations, and unresolved reports may be unavailable to users or support personnel, depending on the existing internal processes.

These points represent potential limitations rather than confirmed deficiencies in the university's current system. Their actual impact requires further investigation.

Need for CampusSignal

CampusSignal is proposed as a centralized platform for reporting and managing Wi-Fi connectivity issues. Campus users would be able to submit structured evidence, while responsible personnel could review issues, assign them for investigation, record interventions, and verify service recovery.

By consolidating incident information and improving status visibility, CampusSignal aims to facilitate communication between students and technical support personnel, make recurring problems easier to identify, and support a more organized incident-resolution process. The system is intended to complement the university's existing IT infrastructure and support services rather than replace the eduroam network itself.

3 Proposed System

3.1 Functional Requirements

The following functional requirements define the connectivity measurement submission and offline queuing. They use the shared CampusSignal vocabulary and state transitions. Priorities are preliminary and should be confirmed during team review.

FR-01 — Submit measurement. The system shall allow an authenticated Campus User to submit a Wi-Fi connectivity measurement or report a failed connection attempt from a specific location on campus.

FR-02 — Capture location and evidence. Upon submission, the system shall capture the user's selected campus location (building, floor, zone), the timestamp of the observation, and the measurement metrics (download speed, upload speed, latency) or the explicit failure status as connectivity evidence.

FR-03 — Evaluate evidence and create issue. When new connectivity evidence is received, the system shall aggregate it with recent data for the selected zone and evaluate it against configured detection thresholds. If the aggregated evidence indicates a fault and no matching active issue exists, the system shall automatically create a candidate Wi-Fi issue in the Detected state for IT review.

FR-04 — Append evidence to active issue. If a newly submitted measurement or failure report matches the location and fault criteria of an already existing active issue, the system shall append this new evidence to the existing issue rather than creating a duplicate.

FR-05 — Retain evidence without creating an issue. If the system determines the transmitted measurement represents acceptable service or does not meet the fault thresholds, it shall record the data for historical and heatmap purposes but shall not create an issue in the Detected state.

FR-06 — Provide submission feedback. The system shall provide the Campus User with a confirmation message upon the successful central recording of their connectivity evidence or failed-connection report.

FR-07 — Offline queue detection. If the system detects that the Campus User has no active internet connection when attempting to submit a report, the system shall locally save the measured evidence or failed-connection report and place it in an offline queue.

FR-08 — Offline queue notification. When a report is placed in the offline queue, the system shall display a notification to the user indicating that the report is locally queued and will be centrally submitted automatically once a connection is established.

FR-09 — Background synchronization. When an active internet connection is restored, the system shall automatically transmit queued connectivity reports in the background without requiring further user intervention, provided the Campus User's consent remains active. If the user withdraws consent before a queued report is transmitted, the system shall delete that unsent report from the local offline queue and shall not transmit it. For transmitted reports, the system shall preserve the original capture time and selected zone while recording the central receipt time separately.

The following functional requirements define the review, triage, decision, and assignment behavior for UC-02. They use the shared CampusSignal vocabulary and state transitions. Priorities are preliminary and should be confirmed during team review.

FR-10 — Issue queue. The system shall provide an issue queue to an authorized IT Administrator, including issues in the Detected state and issues in the Approved state that do not yet have an assigned technician. Each entry shall display the issue identifier, affected zone, submission time, and current state.

FR-11 — Issue evidence. The system shall allow the IT Administrator to inspect the evidence associated with a selected issue, including available connectivity measurements, failed-connection

reports, observation timestamps, the affected zone, and related access-point information. Available evidence shall be distinguishable from missing evidence.

FR-12 — Telemetry and freshness. When AP Controller telemetry is available, the system shall display the relevant access point's reported status, aggregate connected-client count, last-update timestamp, and a freshness indication based on the agreed telemetry-freshness rule. The system shall distinguish an explicitly reported offline access point from telemetry that is stale or unavailable. An unexpected loss of expected telemetry shall be presented as a potential investigation signal, not as conclusive proof of an access-point failure or a user-facing connectivity fault.

FR-13 — Approve issue. For an issue in the Detected state, the system shall allow the IT Administrator to approve the issue and record the decision rationale. The system shall transition the issue from Detected to Approved and record the responsible actor, timestamp, and reason for the transition. Current AP Controller telemetry shall not be a mandatory precondition for approval when the available connectivity evidence and operational context justify action, including a physical inspection. Missing telemetry shall not automatically approve an issue.

FR-14 — Reject issue with reason. The system shall require a non-empty rejection reason before an IT Administrator can reject an issue in the Detected state. After a valid rejection, the system shall transition the issue from Detected to Rejected and preserve the rejection reason, responsible actor, and timestamp.

FR-15 — Assignment eligibility. The system shall permit a technician assignment only when the issue is in the Approved state and has no current assignee. The system shall prevent assignment attempts that violate these conditions.

FR-16 — Select technician and task. For an approved, unassigned issue, the system shall present the IT Administrator with the technicians eligible for assignment and allow the administrator to select one technician. The assignment may request diagnosis or physical inspection when the available evidence suggests a possible access-point, power, or local network-connectivity problem.

FR-17 — Record assignment. When the IT Administrator confirms an assignment, the system shall record the selected technician and transition the issue from Approved to Assigned. The assignment record shall include the responsible administrator, timestamp, and reason or assignment note. The note may specify the requested inspection or diagnostic task.

FR-18 — Notify assignee. After an assignment has been successfully recorded, the system shall notify the selected technician through the system's designated notification mechanism and shall make the assignment outcome available to the IT Administrator.

FR-19 — Retain approved issue when no technician is available. If no eligible technician is available for an approved issue, the system shall retain the issue in the Approved state without an assignee and allow the IT Administrator to return to it for a later assignment attempt. The absence of current telemetry shall not, by itself, prevent assignment when an eligible technician is available and the IT Administrator has justified the need for investigation.

FR-20 — View assigned issues. The system shall allow an authenticated Technician to view the issues currently assigned to them, including the issue identifier, affected zone and current state.

FR-21 — Review assigned issue details. The system shall present the assigned Technician with the selected issue, assignment note, available connectivity evidence and relevant AP information. Telemetry shall carry its last-update information and freshness or availability label; reporter identities shall be hidden in accordance with NFR-19.

FR-22 — Start or resume assigned work. The system shall allow the assigned Technician to start work on an Assigned issue or resume work on a Reopened issue. After a successful recording, the system shall transition the issue to In Progress and record the actor, time and reason.

FR-23 — Enter intervention details. The system shall allow the assigned Technician to enter diagnosis and actions taken for an issue in In Progress.

FR-24 — Validate required intervention information. The system shall require non-empty diagnosis and actions-taken information before accepting an intervention. Missing information shall be identified for correction without creating an accepted intervention or changing the issue state.

FR-25 — Preserve intervention history. The system shall store each accepted intervention with an identifier, the associated issue, the responsible Technician and the recording time. Existing entries shall be read-only; a correction shall be recorded as a new entry, consistent with NFR-16.

FR-26 — Request service verification. The system shall allow the assigned Technician to request verification for an issue in In Progress after the required intervention information has been recorded. The request shall identify the issue and the intervention to be checked.

FR-27 — Record the verification handoff. When the verification request is successfully recorded, the system shall transition the issue from In Progress to Pending Verification, record the actor, time and reason, and make the request available for IT review through UC-04.

FR-28 — Enforce assignment and state rules. The system shall reject an intervention or verification operation when the current actor, assignment or issue state does not permit it. It shall preserve accepted records and explain whether the Technician should return to their issues or review the current state.

FR-29 — Recover from unsuccessful recording. If a work start, intervention or verification request cannot be recorded, the system shall report the unsuccessful operation, preserve previously accepted records and provide a retry or review path. If a previous request has already been accepted, reviewing its result shall not create a second state transition.

FR-30 — Display verification evidence. For an issue in the Pending Verification state, the system shall present the IT Administrator with the newly collected campus user measurements from the affected zone to evaluate service recovery.

FR-31 — Verify service recovery. The system shall permit the IT Administrator to authorize closure only when the available fresh evidence meets the configured acceptance criteria for a successful repair.

FR-32 — Close resolved issue. When the IT Administrator authorizes closure based on sufficient evidence, the system shall transition the issue from Pending Verification to Closed.

FR-33 — Retain pending state for insufficient evidence. If the fresh user measurements do not provide sufficient evidence to confirm recovery, the system shall prevent closure and retain the issue in the Pending Verification state.

FR-34 — Reopen unresolved issue. If the fresh measurements confirm that the connectivity fault persists, the system shall allow the IT Administrator to transition the issue from Pending Verification to Reopened.

FR-35 — Require reason for reopening. The system shall require a non-empty reason from the IT Administrator before an issue can be successfully transitioned to the Reopened state.

FR-36 — Route reopened issue. After an issue is transitioned to Reopened, the system shall retain its existing technician assignment and make the issue available in that technician’s work queue. When

the technician resumes work, the system shall record a transition from Reopened to In Progress with the actor, timestamp, and reason.

FR-37 — Record verification transitions. Whenever an issue transitions from Pending Verification to Closed or Reopened, the system shall record the responsible actor, the exact timestamp, and the reason for the transition.

FR-38 — Update coverage heatmap. Upon transitioning an issue to the Closed state, the system shall immediately update the coverage heatmap to reflect the verified service recovery in the corresponding zone.

FR-39 — Preserve pending issue. If no action is taken or verification cannot be completed in a single session, the system shall retain the issue in the Pending Verification state and allow the IT Administrator to return to it for a later verification attempt.

Functional requirements for UC-05 Investigate Underobserved Zones

FR-40 — Identify underobserved zones. The system shall periodically evaluate mapped campus zones against the configured minimum number of fresh connectivity measurements for the observation period, taking account of each zone's expected-use profile and planned exclusions. It shall flag a zone whose observations are insufficient for its expected use and retain the evaluation time and reason. The evaluation frequency is governed by NFR-12.

FR-41 — Present the investigation list. The system shall allow an authenticated IT Administrator to view flagged zones with their building, floor, zone, observation period, fresh measurement count, expected-use context, flag reason and investigation progress. A flagged zone shall be labelled as having insufficient evidence rather than being classified as good or poor service from the lack of reports alone.

FR-42 — Review zone context and evidence. The system shall present the selected zone's expected-use profile, planned exclusions, available fresh measurements, observation times, related active issues and AP-to-zone mapping. Missing configuration and unavailable evidence shall be identified explicitly so that the IT Administrator can distinguish an evidence gap from an adequately observed zone.

FR-43 — Attempt to retrieve AP telemetry. During zone investigation, the system shall attempt to retrieve relevant AP Controller telemetry through the read-only interface. It shall display the reported AP status, aggregate connected-client count, controller timestamp and freshness or availability label. Stale telemetry shall be excluded from current supporting evidence under NFR-21. Missing telemetry, a low client count or an explicitly reported offline AP shall not, by itself, establish a user-facing connectivity fault.

FR-44 — Request targeted measurements. The system shall allow the IT Administrator to request targeted connectivity measurements for a selected zone, specifying a collection period and instructions. It shall require the zone, a valid collection period and non-empty instructions before accepting the request, and record an identifier, requester and recording time. Requested measurements shall be submitted through UC-01 with the existing consent and zone-level location rules.

FR-45 — Track targeted evidence requests. After accepting a targeted measurement request, the system shall make it available to the intended campus-user recipients and display its recording and delivery outcome to the IT Administrator. It shall link received responses to the request and zone and display collection progress. A delivery failure shall preserve the accepted request and shall not be reported as a recording failure or cause an automatic duplicate request.

FR-46 — Record a justified survey outcome. The system shall allow the IT Administrator to record a survey outcome with a rationale and links to the evidence considered. It shall validate authorization, current investigation context and required outcome information before accepting the record and retain the responsible actor and time. If recording fails, the system shall preserve accepted records, identify the unsuccessful action and provide a retry or review path; it shall not show an unrecorded outcome as accepted.

FR-47 — Link a supported fault to issue management. When fresh connectivity evidence meets the configured fault criteria and the IT Administrator records a fault-supported outcome, the system shall create a linked candidate WiFiIssue in Detected, or attach the relevant evidence and investigation outcome to a matching active issue. The accepted outcome and issue link shall be recorded together. Retrying or reviewing the same accepted outcome shall not create another candidate. UC-05 shall not approve or assign the issue.

FR-48 — Complete an adequately evidenced no-fault investigation. When sufficient fresh connectivity evidence meets the configured acceptable-service criteria, the system shall allow the IT Administrator to record No fault found and complete the investigation with its rationale and evidence references. It shall create no WiFiIssue for that outcome. Completing an investigation shall not close an existing WiFiIssue, whose recovery decision belongs to UC-04.

FR-49 — Retain unresolved investigations. If fresh responses are insufficient, the collection period has not produced adequate evidence, or relevant context is unresolved, the system shall keep the investigation open and state the remaining evidence or configuration gap. It shall allow the IT Administrator to request further measurements or defer review with a recorded reason. Low response, stale telemetry and missing telemetry shall not be converted into a no-fault conclusion or an automatic issue-state transition.

Supporting functional requirements

FR-50 — Secure authentication. The system shall provide a secure login mechanism for all users to authenticate their respective roles (Campus User, IT Administrator, Technician) before accessing role-specific functions, in compliance with NFR-13 and NFR-14.

FR-51 — Heatmap visualization. The system shall provide a campus heatmap interface that visually aggregates and displays Wi-Fi coverage quality based on the collected connectivity evidence and known access-point data.

FR-52 — Heatmap status differentiation. The system shall visually distinguish different coverage states on the heatmap (good, fair, poor, insufficient evidence, and planned exclusions) using standardized text labels or icons in addition to color indicators, ensuring status is not conveyed by color alone.

FR-53 — Access point mapping. The system shall maintain an access point (AP) to zone mapping, configurable by the IT Administrator, to correlate AP telemetry with user-reported locations.

FR-54 — Expected-use profiling. The system shall store an expected-use profile for each campus zone, allowing the IT Administrator to define the expected usage level and typical hours, to differentiate between naturally low usage and an abnormal lack of connectivity reports.

FR-55 — Account registration. The system shall allow Campus Users to self-register for an account using their university email address, automatically assigning them the Campus User role and preventing the selection of Technician or IT Administrator roles.

FR-56 — Informed consent. The system shall present a privacy notice to the Campus User and explicitly record their consent before allowing the first connectivity measurement. If consent is withdrawn, the system shall prevent new measurements from being started or collected and

delete any connectivity evidence that has not yet been transmitted and remains in the local offline queue.

FR-57 — View own submissions. The system shall allow the Campus User to view a history of their own queued and centrally submitted connectivity reports, their submission status, and the status of any linked Wi-Fi issue.

FR-58 — Configure thresholds and mappings. The system shall allow authorized IT Administrators to configure issue detection thresholds, the measurement window, verification criteria, AP-to-zone mappings, and planned exclusions, in accordance with NFR-23.

FR-59 — Retrieve AP telemetry. The system shall periodically retrieve per-AP online status, aggregate connected-client counts, and the Controller timestamp from the AP Controller via a read-only interface, in accordance with NFR-20 and NFR-21. The system shall not present assumed data when the AP Controller is unreachable.

3.2 Nonfunctional Requirements

The nonfunctional requirements below constrain how CampusSignal delivers the functional requirements in Section 3.1. Each requirement has a stable identifier, a verifiable fit criterion, and its main source. Values described as configurable can be changed by IT administrators (NFR-23). Because implementation and deployment are outside the course scope, some fit criteria describe how the requirement would be verified on the navigable prototype and others how it would be verified on a deployed system.

3.2.1 Usability and Accessibility

NFR-01 — Quick evidence submission. From the campus-user home screen, a campus user shall be able to start a connectivity test, review its result, and submit it as evidence in no more than three interactions.
Verification: Walk through the UC-01 main flow in the prototype and count the interactions; the automatic test run is not counted.

NFR-02 — Readable contrast. All screens shall meet WCAG 2.1 Level AA contrast: at least 4.5:1 for normal text and at least 3:1 for large text, icons, and status indicators [W3C 2018].
Verification: Check every text and status color pair in the mock-ups with a contrast checker.

NFR-03 — Status not shown by color alone. Heatmap zone categories (good, fair, poor, insufficient evidence, planned exclusion) and WiFiIssue states shall each be identifiable by a text label, icon, or pattern in addition to color.
Verification: View the heatmap and issue-list screens in grayscale; every category and state must still be distinguishable.

NFR-04 — Explained outcomes. Every rejection, error, waiting, or pending state shown to a user shall state the reason and the next action available to that user.
Verification: Review all error and recovery screens of the prototype against this rule.

3.2.2 Reliability and Availability

NFR-05 — Service availability. Evidence submission, heatmap viewing, and issue management shall be available at least 99% of the time between 07:00 and 23:00 each day, measured per calendar

month and excluding maintenance announced at least 48 hours in advance. *Verification: Monthly availability report on a deployed system (acceptance target).*

NFR-06 — No loss of confirmed data. Once the system confirms an evidence submission or a WiFiIssue state transition to a user, that record shall not be lost if the system restarts. *Verification: Restart the system after a confirmed submission and a confirmed transition; both records must be present with unchanged content.*

NFR-07 — Offline evidence queue. If evidence cannot be sent because the device has no usable connection, the campus-user client shall keep it on the device for at least 24 hours, including across application restarts, and shall send it automatically within 1 minute after a connection becomes available, provided consent remains active. If the user withdraws consent before transmission, the client shall delete the unsent evidence from the local queue and shall not transmit it. For transmitted evidence, the client shall preserve the original capture time and zone. Verification: Submit a test result in airplane mode, restart the application, reconnect while consent remains active, and check that the evidence arrives with its original capture time and zone. Repeat the test, withdraw consent before reconnecting, and confirm that the queued evidence is deleted and is not transmitted.

NFR-08 — Operation without current AP telemetry. If the AP Controller cannot be reached, or the telemetry of an AP is stale (NFR-21), the system shall continue to accept user evidence and show the heatmap from user measurements. Affected AP data shall be labelled “unavailable” or “stale” together with the time of its last successful update and shall never be presented as current. *Verification: Run the UC-01 and heatmap scenarios with the controller disconnected; both must complete and every affected screen must show the label and last-update time.*

3.2.3 Performance

NFR-09 — Connectivity test cost. A single connectivity test shall complete within 20 seconds and transfer no more than 10 MB of data. *Verification: Run repeated tests against the test endpoint (NFR-22) and record duration and transferred data.*

NFR-10 — Response time under load. With up to 500 concurrent users, the system shall confirm receipt of submitted evidence within 3 seconds and display one building floor of the heatmap within 3 seconds for 95% of requests made over the campus network. *Verification: Load test with 500 simulated users; measure the 95th-percentile response times.*

NFR-11 — Detection delay. The system shall evaluate new evidence against the detection thresholds and either create a candidate issue or attach the evidence to a matching active issue within 2 minutes of receipt. *Verification: Compare evidence receipt times with issue creation or attachment times in test data.*

NFR-12 — Underobserved-zone evaluation frequency. The system shall re-evaluate every mapped zone for underobservation at least once every 24 hours and add newly flagged zones to the IT administrator’s list in the same evaluation. *Verification: Check the evaluation times of all zones over a simulated 7-day period.*

3.2.4 Security

NFR-13 — Authenticated access. Every function except sign-in, account registration, and the privacy notice shall require an authenticated user. Self-registration shall accept only university e-mail addresses and shall always create a campus-user account; the technician and IT administrator roles can be assigned only by an IT administrator. Sessions of IT administrators and technicians shall end after 30 minutes of inactivity.
Verification: Attempt each function without signing in; try to register with a non-university e-mail address; check that registration never offers a staff role; leave a staff session idle for 30 minutes.

NFR-14 — Role-based access. The system shall restrict functions by role. Campus users submit evidence, view the heatmap and their own submissions, and answer measurement requests. Technicians view and update only the issues assigned to them. Only IT administrators triage, assign, close, and reopen issues and change configuration.
Verification: For each role, attempt every function reserved for another role; all attempts must be denied.

NFR-15 — Encrypted communication. All communication between clients, the system, and the AP Controller shall be encrypted with TLS 1.2 or later.
Verification: Interface specification review and inspection of network traffic.

NFR-16 — Append-only history. Recorded WiFiIssue transitions (actor, time, reason) and intervention records shall be append-only: no role, including IT administrators, can edit or delete an existing entry, and corrections are added as new entries.
Verification: Attempt to edit and delete history entries with each role; all attempts must be denied.

3.2.5 Privacy, Consent, and Legal Obligations

NFR-17 — Informed consent. Before a campus user's first measurement, the system shall show a privacy notice and record the user's explicit consent. The user shall be able to withdraw consent at any time, after which no new measurements are collected from that user. Processing of personal data shall comply with Law No. 6698 on the Protection of Personal Data (KVKK).
Verification: Follow the consent path in the prototype; withdraw consent and confirm that a new test cannot be started.

NFR-18 — Data minimization. The system shall record a measurement's location only at zone level (building, floor, zone) and shall not collect continuous location traces, browsing history, application usage, or traffic content.
Verification: Compare the fields of a measurement record with this list; none of the excluded data may appear.

NFR-19 — Identity visibility and retention. Submitter identities shall be visible only to IT administrators; technicians, other campus users, and the heatmap shall show evidence without identity. The link between a measurement and its submitter shall be removed after 180 days, while the anonymous measurement values and the issue history are kept.
Verification: Open the same evidence with each role; inspect records older than 180 days for submitter links.

3.2.6 Interface Constraints

NFR-20 — Read-only AP Controller interface. The system shall only read data from the AP Controller: per-AP online status, aggregate connected-client count, and the controller's timestamp for these values. It shall not send configuration, restart, or any other write command to the controller, and it shall not request per-client identifiers such as MAC addresses, device names, or usernames.
Verification: Review the interface specification; every controller operation used must be a read of the three listed values.

NFR-21 — Telemetry update interval and freshness threshold. The system shall request telemetry for every mapped AP at least every 5 minutes. Telemetry whose controller timestamp is more than 15 minutes old (three missed updates) is stale: it shall be displayed only with a "stale" label and its age, and it shall not be counted as supporting evidence in issue detection, verification, or underobserved-zone evaluation. The threshold is configurable between 5 and 60 minutes. *Verification: Feed simulated telemetry that is 14 and 16 minutes old; the first must be treated as fresh, the second as stale and excluded from evidence.*

NFR-22 — Controlled test endpoint. All speed and latency tests shall be run against a university-controlled test endpoint inside the campus network, so that results from different zones are comparable and reflect campus Wi-Fi rather than external Internet conditions. *Verification: Interface specification review.*

3.2.7 Supportability and Implementation Constraints

NFR-23 — Configuration without software changes. IT administrators shall be able to change detection thresholds, verification acceptance criteria, the telemetry freshness threshold, AP-to-zone mappings, expected-use profiles, and planned exclusions through the system without software changes. Each change shall take effect within 5 minutes and be recorded with actor and time. *Verification: Change a threshold through the configuration screen and observe the new behavior and the change record.*

NFR-24 — Target platforms. Campus-user and technician functions shall run on smartphones with the current and previous major versions of Android and iOS. IT administrator functions shall run in the current versions of Chrome, Firefox, Edge, and Safari on desktop screens of at least 1280 × 720 pixels.
Verification: The prototype contains mobile frames for campus users and technicians and desktop frames for IT administrators; compatibility testing on a deployed system.

3.2.8 Assumptions

A-01 — The university grants read-only API access to the AP Controller, which supplies per-AP online status, aggregate connected-client counts, and timestamps. This assumption from the approved proposal must be confirmed with University IT; until then, scenarios and mock-ups use simulated telemetry with the same content.

A-02 — A university-controlled test endpoint (NFR-22) can be hosted inside the campus network.

A-03 — CampusSignal is specified as a standalone application with its own sign-in and registration. Integration into the university's existing MCBU campus mobile app is considered as a possible later

deployment option; in that case sign-in and account management would be provided by the host app. This option is outside the scope of this RAD, and the integration approach will be evaluated in the SDD. The requirements above do not depend on either deployment option.

Consent withdrawal and offline queue: If a Campus User withdraws consent while unsubmitted evidence remains in the offline queue, the system deletes that unsent evidence locally and does not transmit it. FR-09, FR-56, and NFR-07 define this behavior.

References

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3.3 System Models

This RAD-v1 report consolidates SC-01–SC-05 and the detailed descriptions of UC-01–UC-05, together with the top-level use-case diagram, screen specifications, and traceability mapping. The Object Model and Dynamic Models are addressed in RAD-v2.

3.3.1 Scenarios

SC-01 — Submit Connectivity Evidence (Offline Recovery)

Context: A student, Elif, attempts to study in a campus library study area but experiences repeated Wi-Fi connectivity drops. She decides to submit connectivity evidence through CampusSignal. However, during the submission, her device lacks an active internet connection. The system queues the report locally and automatically submits it when she reconnects to a cellular network.

Actors: Campus User (Elif Yılmaz).

Trigger: The Campus User initiates a connectivity measurement using the CampusSignal application from their current location.

Preconditions: The Campus User has an authenticated session and has explicitly granted consent for data collection; the CampusSignal application is running on their mobile device.

Main success scenario

1. Elif selects her current location (building, floor, and zone) in the CampusSignal application and requests to submit connectivity evidence.

2. The application attempts to perform a network measurement against the university-controlled test endpoint but detects that there is no active internet connection to reach it.

3. The system captures the selected location, the timestamp, and the failed connection status as connectivity evidence.

4. The system locally stores the connectivity evidence in an offline queue and displays a notification to Elif indicating that the report is saved and will be submitted automatically when a connection is established, provided consent remains active.

5. Elif leaves the library and her device automatically connects to her cellular data network.

6. The CampusSignal application detects the active internet connection and automatically transmits the queued connectivity evidence in the background.

7. The system aggregates the received evidence with recent measurements for that zone. Because the total number of poor measurements now meets the configured detection thresholds, the system identifies a fault and creates a candidate Wi-Fi issue in the Detected state.

8. The system notifies Elif that her queued report has been submitted.

Alternative and exception scenarios

SC-01-A — Active internet connection available: If Elif has a weak but active connection, the system successfully performs the measurement, transmits the evidence immediately, aggregates it to check thresholds, and displays a success message without utilizing the offline queue.

SC-01-B — Match with an active issue: When the system evaluates the transmitted evidence (either queued or immediate), if it matches the location and fault criteria of an existing active issue, the system appends the new evidence to the existing issue rather than creating a duplicate in the Detected state.

Outcome: On success, the connectivity evidence is recorded centrally, the user receives a confirmation, and either a new candidate issue is created (Detected) or an existing issue is updated. If the connection cannot be established at all, the evidence remains safely stored in the local offline queue.

SC-02 — Triage and Assign an Issue

Context: A student, Elif, experiences repeated Wi-Fi connectivity problems in a campus library study area and submits connectivity evidence through CampusSignal. A candidate issue is created in the Detected state. Current telemetry for the related access point cannot be retrieved. IT Administrator Selin reviews the evidence and determines that the access point or its local connectivity should be physically inspected. Technician Murat is available for assignment.

Actors: IT Administrator (Selin), Technician (Murat). The AP Controller is an external source of network telemetry.

Trigger: Selin begins reviewing the detected issue in the issue queue.

Preconditions: Selin has an authenticated and authorized IT Administrator session; the candidate issue is in Detected and has at least one linked connectivity report or measurement; stored evidence is available. AP telemetry may be current, stale, explicitly report an access point as offline, or be unavailable.

Main success scenario

1. Selin requests the current issue queue. The system lists Detected issues and Approved issues that do not yet have an assignee.

2. Selin selects the issue associated with Elif's report. The system presents the issue identifier, affected zone, evidence, observation times, and related access-point information.

3. The system displays available AP Controller status and aggregate client-count information, including the last-update time and freshness indication. In this scenario, current telemetry is unavailable; the system identifies that limitation rather than automatically labelling the AP as offline.

4. Selin considers the repeated connectivity evidence and the unexpected absence of expected telemetry. She concludes that the available evidence justifies investigation and records an approval rationale requesting physical inspection of the access point and its local connectivity.

5. The system records the decision and transitions the issue from Detected to Approved.

6. The system lists eligible technicians. Selin selects Murat and adds an assignment note requesting a physical inspection and diagnosis of the access point and local connectivity.

7. The system validates that the issue is Approved and unassigned and that Murat is eligible. Selin confirms the assignment.

8. The system records the assignee, responsible administrator, timestamp, and assignment note. It transitions the issue from Approved to Assigned, attempts to notify Murat, and presents the result to Selin.

9. Murat uses the assignment details to investigate the problem. He will record his findings and actions through UC-03, Record an Intervention. Service recovery can later be assessed through UC-04, Verify Service Recovery.

Alternative and exception scenarios

SC-02-A — Reject the candidate issue: If Selin determines that the evidence does not justify accepting the issue, she supplies a non-empty rejection reason. The system records the reason, actor, and timestamp, transitions Detected to Rejected, and ends the use case without assignment.

SC-02-B — Stale or unavailable telemetry: The system marks telemetry as stale or unavailable and shows its last-update information when available. Selin may still approve the issue and request inspection if the remaining evidence and operational context justify action. Otherwise, the issue remains in its current state while further evidence is sought. Missing telemetry is neither proof of a connectivity fault nor a reason that automatically blocks a justified assignment.

SC-02-C — Explicitly reported offline AP: If the AP Controller reports the AP as offline, the system displays the reported status and timestamp. Selin may use this as supporting evidence to approve the issue and assign a technician. The reported status does not establish the exact physical cause.

SC-02-D — No eligible technician: If no technician is available, the system retains the issue in Approved without an assignee, informs Selin, and keeps the issue available for a later assignment attempt.

SC-02-E — Assignment recording fails: If the system cannot store the assignment, it reports the failure and preserves the issue in Approved without an assignee. It does not report an assignment as successful.

SC-02-F — Notification fails: If the assignment is stored but notification delivery fails, the issue remains Assigned. The system indicates the notification failure without undoing the stored assignment.

Outcome: On success, the issue is Assigned and the assignee, transition history, and assignment note are recorded. On rejection, it remains Rejected with a reason. If no technician is available or assignment storage fails, it remains Approved and unassigned.

SC-03 — Record an Intervention

Murat is a campus Technician. Issue CS-024 concerns repeated connection failures in the Library, Floor 2, and is mapped to AP-LIB-04. IT has approved the issue and assigned it to Murat at 14:17 with a note requesting an inspection of the AP, its power and local connection. The AP Controller has not supplied a current response; the last successful response was at 14:05. This fictional case continues the issue shown in the shared prototype.

Actors: Technician, represented by Murat. The IT Administrator performs the later UC-04 review.

Trigger: Murat opens CS-024 from his assigned issues and requests to start work.

Preconditions: Murat has an authenticated Technician session. CS-024 exists in Assigned, is assigned to Murat, and its stored assignment and evidence can be read. Current AP Controller telemetry is not required to record a justified inspection.

1 Murat opens CS-024. CampusSignal presents the Library Floor 2 location, AP-LIB-04 mapping, the assignment note and three failed connection reports observed at 14:02, 14:06 and 14:10. Reporter identities are hidden.
2 CampusSignal labels AP telemetry Unavailable and shows the last response time, 14:05. It does not infer that the AP itself is offline.
3 Murat confirms that he is starting the assigned inspection. At 14:21, CampusSignal records the actor and reason and transitions Assigned to In Progress.
4 During the physical inspection, Murat finds a loose uplink cable. He reseats it and checks the link indicator. These physical actions take place outside CampusSignal; the application records his account of them.
5 Murat enters the diagnosis and actions taken. CampusSignal validates both required fields and, at 14:37, records intervention INT-024-01 linked to CS-024 and Murat. The issue remains In Progress.
6 Murat requests service verification. At 14:38, CampusSignal records the request and the transition to Pending Verification. It makes the request and intervention available for the IT review.
7 Murat sees confirmation of the recorded request. His issues list now shows CS-024 as Pending Verification. The IT Administrator can assess fresh measurements through UC-04.

CS-024 is Pending Verification. The intervention, request and actor-time-reason history are recorded. No closure or successful recovery is asserted by this use case.

SC-03-A Missing diagnosis

If Murat tries to save without diagnosis or actions, CampusSignal identifies the missing fields. No intervention is accepted; he corrects the form and submits it again.

SC-03-B Request not recorded

If the intervention is saved but the verification request cannot be recorded, INT-024-01 remains available and CS-024 remains In Progress. Murat can retry the request or leave it for later.

SC-03-C Stale telemetry

If the controller returns a reading last updated at 13:58 during a 14:20 review, CampusSignal labels it Stale, shows its age of 22 minutes and excludes it from current supporting evidence. The inspection and recording can continue.

SC-04 — Verify Service Recovery

Context: Technician Murat has completed an intervention for a Wi-Fi issue in a campus library study area and requested verification. The issue is now in the Pending Verification state. To ensure the problem is actually solved, the system has been collecting fresh Wi-Fi connectivity measurements

from students in that zone. IT Administrator Selin reviews this fresh evidence. Based on the data, she will either close the issue, reopen it if the fault persists, or wait if the evidence is insufficient.

Actors: IT Administrator (Selin).

Trigger: Selin selects an issue currently in the Pending Verification state from the issue queue to review its recovery status.

Preconditions: Selin has an authenticated and authorized IT Administrator session; the issue is in the Pending Verification state; a technician has previously recorded an intervention for this issue.

Main success scenario

1. Selin requests to view the details of the issue currently in the Pending Verification state.

2. The system presents the issue identifier, affected zone, intervention history recorded by Murat, and the fresh Wi-Fi measurement evidence gathered from campus users since the intervention.

3. Selin evaluates the quantity, freshness, relevance and quality of the measurements against the configured evidence-sufficiency and service-recovery criteria. In this success scenario, the evidence is sufficient and supports recovery.

4. Selin authorizes the closure of the issue.

5. The system records the closure decision, reason, responsible actor (Selin) and exact timestamp.

6. The system transitions the issue state from Pending Verification to Closed.

7. The system immediately updates the coverage heatmap to reflect the verified service recovery (green) for that zone.

Alternative and exception scenarios

SC-04-A — Reopen unresolved issue: If fresh measurements indicate that poor connectivity persists (for example, users still experience connection drops), Selin reopens the issue and supplies a mandatory, non-empty reason. The system records the reason, actor and timestamp, transitions the issue from Pending Verification to Reopened, retains Murat as the assignee, and makes the issue visible in his work queue. When Murat resumes work, the system records Reopened → In Progress with the actor, timestamp and reason.

SC-04-B — Insufficient evidence: If the system displays that not enough fresh measurements have been collected to evaluate recovery, Selin cannot authorize closure. The system retains the issue safely in the Pending Verification state without further action, waiting for more campus users to submit data.

Outcome: On success, the issue is Closed, the coverage heatmap reflects verified recovery, and the transition history is recorded. If the fault persists, the issue becomes Reopened with a recorded reason, retains its existing technician assignment, and appears in the technician's work queue. When the technician resumes work, the state changes to In Progress. If evidence is insufficient, the issue remains Pending Verification.

SC-05 — Investigate Underobserved Zones

IT Administrator Selin reviews a fictional Engineering Building study zone on Floor 1, Zone C, mapped to AP-ENG-03. The expected-use profile indicates weekday teaching use, but the current observation period contains only two fresh measurements against a configured minimum of five. The zone is not a planned exclusion. CampusSignal has flagged insufficient evidence. These identifiers and values illustrate the scenario rather than assert live campus conditions.

1. At 10:00, Selin opens the underobserved-zone list and selects Engineering Building, Floor 1, Zone C. CampusSignal shows the observation period, two fresh measurements, expected weekday use and the reason for the flag.

2. CampusSignal attempts to retrieve AP-ENG-03 telemetry. The controller does not respond, so the system displays Unavailable and the last successful update when known. Selin does not infer that the AP is offline or that the zone is empty.

3. Selin requests targeted measurements for the next teaching period and asks participating campus users to select Zone C and submit a connectivity test or failed-connection report. CampusSignal records request TMR-005-01 and makes it available to the intended recipients.

4. Five campus users later submit fresh evidence through UC-01 after the existing consent checks. Three tests fail to connect and two show poor service. CampusSignal links the responses to TMR-005-01 and updates the collection progress.

5. Selin resumes the investigation. The five responses satisfy this example's configured evidence and fault criteria. She records Fault supported with a rationale referring to the failed tests and poor results.

6. CampusSignal validates the current evidence and finds no matching active issue. It records survey outcome SUR-005-01 and creates a linked candidate in Detected. It confirms the outcome and issue link to Selin.

7. Selin can follow the candidate into UC-02 for a reasoned triage decision. The candidate has not been approved or assigned by UC-05.

SC-05-A Adequate normal service. If the fresh measurements meet the configured acceptable-service criteria, Selin records No fault found with the evidence references. The investigation is completed and no new WiFiIssue is created.

SC-05-B Insufficient responses. If only two eligible responses arrive, the zone remains insufficiently observed. Selin records the evidence gap and requests further measurements or defers review. The absence of reports is not treated as proof of good coverage.

SC-05-C Matching active issue. If the supported fault matches an active issue, the system links the survey and new evidence to that issue instead of creating another candidate. The existing issue keeps its current state.

SC-05-D Planned exclusion. If Selin confirms a configured planned exclusion for the relevant period, she records that context and the reason for ending the investigation. The zone remains labelled Planned exclusion, and is not labelled good coverage.

3.3.2 Use Case Model

Figure 1 summarizes the five critical use cases and supporting functions for account registration, coverage exploration, configuration, and telemetry. The Scheduler is internal to CampusSignal and is not an external actor. UC-02 and UC-05 include an attempt to retrieve AP Controller telemetry; the result may be current, stale, explicitly offline, or unavailable, and missing telemetry alone does not establish an AP fault.

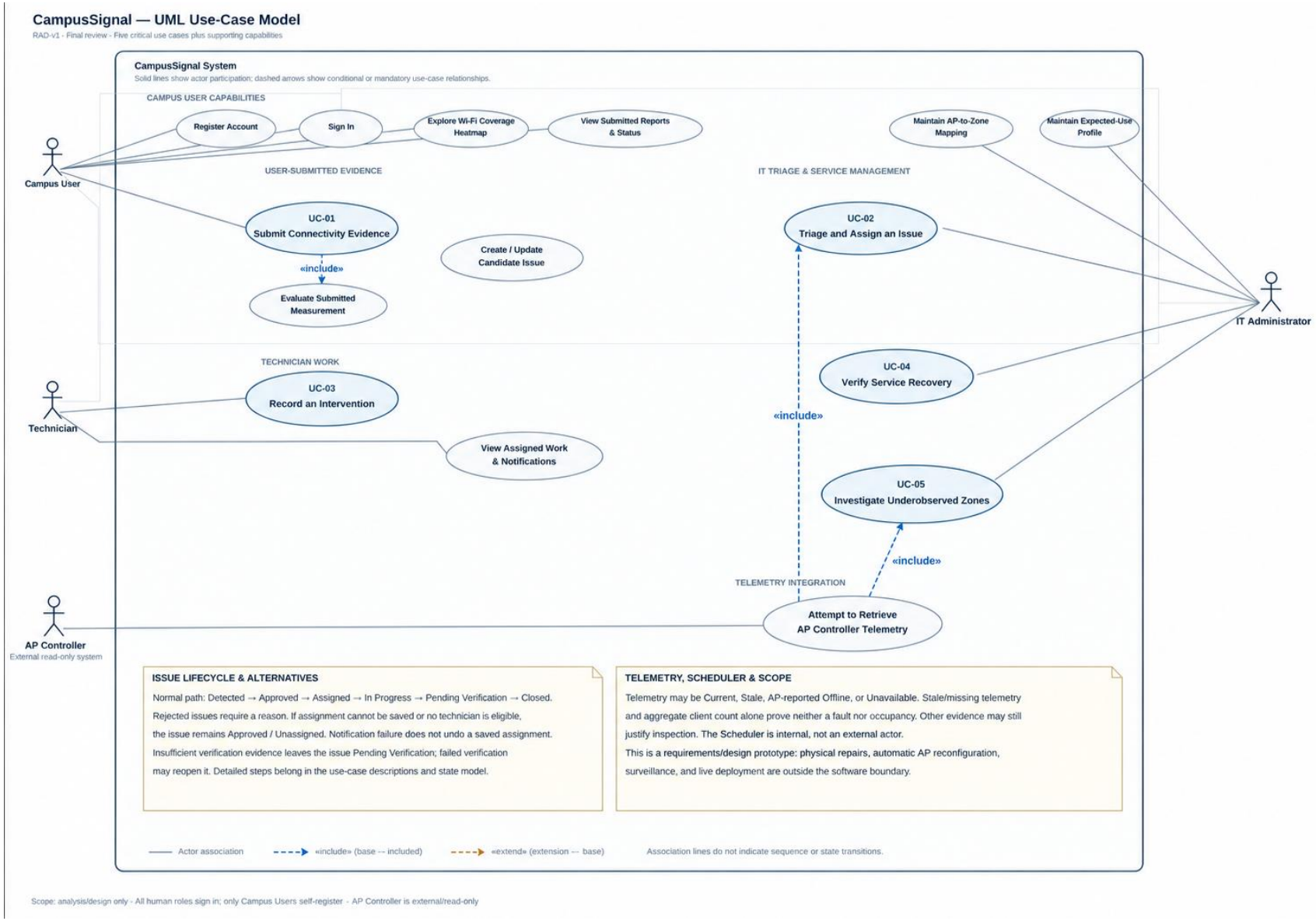


Figure 1. CampusSignal top-level UML use-case diagram supplied by Efe. The original diagram is retained; detailed main and exception flows are specified below.

UC-01 — Submit Connectivity Evidence

Primary actor: Campus User.

Supporting actors: None.

Trigger: The Campus User requests to submit a Wi-Fi connectivity measurement or report a failed connection from a specific campus location.

Preconditions: The Campus User has an authenticated session and has explicitly granted consent for data collection. The CampusSignal application is running on the user's mobile device.

Success postconditions: The connectivity evidence is centrally recorded. Depending on configured thresholds, a candidate Wi-Fi issue is created, evidence is appended to an existing issue, or the measurement is simply logged. The Campus User receives a confirmation.

Alternative postconditions: If the device lacks an active internet connection, the evidence is stored in a local offline queue while consent remains active. If consent is withdrawn before transmission, the queued evidence is deleted and is not transmitted.

Main success flow

1. The Campus User selects their current location (building, floor, and zone) in the application and requests to submit connectivity evidence.

2. The application successfully performs a network measurement against the campus test endpoint.

3. The application transmits the evidence (metrics, selected location, and timestamp) to the central system.

4. The system aggregates the received evidence with recent data for that zone and evaluates it against configured detection thresholds.

5. The system confirms the aggregated evidence justifies a fault and creates a new candidate Wi-Fi issue in the Detected state.

6. The system records the connectivity evidence and links it to the newly created issue.

7. The system displays a confirmation message to the Campus User indicating that the report has been successfully submitted.

Alternative and exception flows

A1 — Offline queue (No active connection before test): At step 2, if the application detects that the device has no active internet connection, it captures the selected location, timestamp, and failed status, and saves this evidence locally in an offline queue. The application notifies the user that the report is saved and will be submitted automatically. The use case is suspended.

A2 — Offline queue (Connection lost during transmission): After step 3, if the transmission to the central system fails due to a sudden loss of connection, the application saves the measured evidence

locally in an offline queue and notifies the user that it will be submitted automatically when the connection is restored. The use case is suspended.

A3 — Background synchronization: Following A1 or A2, when the application detects that an active internet connection has been established, it automatically transmits the queued evidence in the background. The system resumes at step 4.

A4 — Match with an existing active issue: At step 5, if the system evaluates the evidence and determines it matches an existing active issue, it appends the new evidence to the existing issue rather than creating a duplicate. The system resumes at step 7.

A5 — Evidence does not meet fault criteria: At step 5, if the aggregated evidence represents acceptable service or does not yet meet configured detection thresholds, it records the measurement for heatmap aggregation but does not create an issue in Detected. The system resumes at step 7.

A6 — Explicit failed connection report: At step 1, if the Campus User already knows the connection is completely dead and the test cannot run, they can choose to explicitly submit a "Failed Connection" report manually for their selected zone without waiting for the speed test to time out. The system accepts this as negative evidence and resumes at step 3.

A7 — Consent withdrawn: At any point, if the system detects that the user has withdrawn their privacy consent, the system blocks the measurement submission, deletes any evidence that has not yet been transmitted and remains in the local offline queue, prevents any new measurement from starting, and redirects the user to the privacy settings.

A8 — Measurement timeout: At step 2, if a measurement takes longer than 20 seconds, the system records it as a failed connection attempt and ends the measurement, resuming at step 3.

Related functional requirements: FR-01 through FR-09, FR-55, FR-56, FR-57.

UC-02 — Triage and Assign an Issue

Primary actor: IT Administrator.

Supporting actors: Technician, who receives the assignment and carries out the requested diagnosis or inspection; AP Controller, an external source of reported access-point status and aggregate connected-client counts.

Trigger: The IT Administrator opens an issue in Detected for review, or returns to an Approved issue without an assignee to complete assignment.

Preconditions: The IT Administrator has an authenticated, authorized session. An issue exists in Detected or is already Approved without an assignee. Any stored issue evidence is available. The system attempts to retrieve relevant AP Controller telemetry during triage; the response may be current, stale, explicitly report an AP as offline, or be unavailable. Current telemetry is not a precondition for a justified decision.

Success postconditions: The issue is Assigned to a technician. Approval and assignment transitions, actors, timestamps, rationale, and assignment note are recorded. The system has attempted to notify the assigned technician.

Alternative postconditions: A rejected issue is Rejected with a recorded reason. If no eligible technician is available, the issue remains Approved without an assignee. If assignment storage fails, the previous state is preserved.

Main success flow

1. The IT Administrator requests the issue queue.

2. The system lists issues in Detected and approved issues in Approved that have no assignee.

3. The IT Administrator selects an issue.

4. The system presents the issue identifier, affected zone, submitted reports, available measurements, observation timestamps, and related access-point information.

5. The system displays available AP Controller status and aggregate client counts with their last-update timestamp and freshness indication. It distinguishes an explicitly reported offline AP from stale or unavailable telemetry.

6. If the issue is Detected, the IT Administrator reviews the available evidence, approves the issue, and records a rationale. The system records the decision and transitions the issue to Approved. An issue already in Approved skips this decision step.

7. The system presents eligible technicians. The IT Administrator selects a technician and may enter an assignment note, such as a request to physically inspect an AP or check local power and connectivity when expected telemetry is unavailable.

8. The system validates that the issue is Approved, has no current assignee, and is assignable to the selected technician.

9. The IT Administrator confirms the assignment.

10. The system records the assignee, responsible administrator, timestamp, and assignment reason or note. It transitions Approved to Assigned.

11. The system attempts to notify the technician and displays the assignment outcome to the IT Administrator.

Alternative and exception flows

A1 — Reject issue: The IT Administrator may reject only a Detected issue and must provide a non-empty reason. The system records the reason, actor, and timestamp, transitions Detected to Rejected, and ends without assignment.

A2 — Missing rejection reason: The system refuses to record the rejection and requests a reason. The issue remains Detected; the administrator can enter a reason or choose another decision.

A3 — Stale or unavailable telemetry: The system marks telemetry stale or unavailable, showing the last-update information when available. It does not automatically classify the AP as offline. The IT Administrator may proceed when other evidence and operational context justify investigation; otherwise, the issue remains in its current state while more evidence is sought. An underobserved-zone concern may be followed up through UC-05.

A4 — AP explicitly reported offline: The system presents the status reported by the AP Controller and its timestamp. The IT Administrator may consider it as supporting evidence and request physical inspection; the status does not establish the exact physical cause.

A5 — No eligible technician: The system informs the IT Administrator that no assignment can be completed. The issue remains Approved without an assignee and can be revisited later.

A6 — Assignment validation or storage failure: The system refuses an invalid assignment or reports that it could not store the assignment. It preserves the issue's prior state and does not report a successful assignment.

A7 — Notification delivery failure: If the assignment has already been stored, the issue remains Assigned. The system indicates the delivery problem without undoing the assignment record.

A8 — Unauthorized access: The system denies access to review and assignment functions and performs no issue-state or assignment change.

A9 — Issue already Approved and unassigned: If the use case starts from an Approved issue awaiting assignment, the system skips the approval decision in Step 6 and continues at Step 7. It does not record a duplicate approval transition.

Related functional requirements: FR-10 through FR-19.

UC-03 — Record an Intervention

Primary actor: Technician.

Supporting actors: None directly required in this interaction. The IT Administrator performs the later UC-04 review.

Trigger: The Technician requests to start or continue work on an issue assigned to them.

Preconditions: The Technician has an authenticated, authorized session. The issue exists, is assigned to this Technician, and is in Assigned, Reopened, or In Progress. Stored issue details are readable; current telemetry is not required.

Success postconditions: An accepted intervention and verification request exist; the issue is Pending Verification; actor, time and reason are recorded for transitions; UC-04 can review the recorded request.

Other outcomes: Cancellation preserves the issue's previous state, whether Assigned, Reopened or In Progress. Failed recording preserves previously accepted data. Unauthorized or stale operations do not change accepted records.

Main success flow

1. The Technician requests the details of an issue assigned to them.
2. The system checks access and presents the issue identifier, zone, state, assignment note, available evidence and relevant AP information. It labels stale or unavailable telemetry and hides reporter identities.
3. For an Assigned or Reopened issue, the Technician requests to start or resume work and confirms the reason.
4. The system checks the current assignment and state, records the start or resumption with actor, time and reason, and transitions Assigned or Reopened to In Progress. If the issue is already In Progress, it does not create another start transition.
5. The Technician supplies diagnosis and actions taken for the intervention and submits those details.
6. The system checks authorization, the current state and both required fields. It records an intervention linked to the issue, Technician and recording time and confirms the result. The issue remains In Progress.
7. The Technician requests verification of the recorded intervention.
8. The system checks that the issue is In Progress, belongs to the Technician and has the required recorded intervention information.

9. The system records the verification request and the transition to Pending Verification with actor, time and reason, and makes the request available for UC-04.
10. The system confirms the recorded request and presents the resulting state to the Technician.

A1 — Work is already In Progress

Step 3. The system omits the start transition and continues at Step 5. No second start transition is created.

Result: In Progress. UI: UI-21A; UI-24A.

A2 — Technician cancels before starting

Step 3. The system returns to the issue without recording a work start. The use case ends for this attempt.

Result: The previous Assigned or Reopened state is preserved. UI: UI-20; UI-21 illustrate the Assigned case; resuming a Reopened issue follows the same specified work-start rule.

A3 — Required information is missing

Step 6. The system identifies the missing diagnosis or actions and refuses to accept the entry. The Technician corrects it and resumes at Step 5, or cancels.

Result: In Progress; no new intervention. UI: UI-22B.

A4 — Actor or assignment does not permit the operation

Steps 2, 4, 6 or 8. The system denies the operation without changing accepted data and directs the Technician to their current issues. The attempted operation ends.

Result: Previous accepted state. UI: UI-20C.

A5 — Work start cannot be recorded

Step 4. The system shows that work has not been recorded as started. The Technician retries Step 3 or returns to the issue.

Result: The previous Assigned or Reopened state is preserved. UI: UI-21B illustrates the Assigned work-start failure.

A6 — Intervention cannot be recorded

Step 6. The system does not issue an intervention receipt. It retains entered values for the current form view and allows correction or a retry of Step 5. No offline persistence is promised for technician forms.

Result: In Progress; previous entries retained. UI: UI-22C.

A7 — Verification request cannot be recorded

Step 9. The system retains the accepted intervention and explains that verification has not started. The Technician retries Step 7 or defers the request.

Result: In Progress; intervention retained. UI: UI-23B.

A8 Telemetry is stale

Step 2. The system displays the stale label, update time and age. It excludes the reading from current supporting evidence, presents the available reports and continues at Step 3 or Step 5.

Result: Unchanged by telemetry limitation. UI: UI-20B.

A9 AP Controller cannot be reached

Step 2. The system displays Unavailable and the last successful update if known. It treats the AP status as unknown and continues with available reports and assignment context.

Result: Unchanged by telemetry limitation. UI: UI-20.

A10 Technician defers verification

After Step 6. The system preserves the accepted intervention. The Technician can return to the saved history and request verification later at Step 7. This attempt ends before verification is requested.

Result: In Progress; intervention retained. UI: UI-20D; UI-24B.

A11 Issue state changed or previous request is already recorded

Steps 4, 6, 8 or 9. The system refuses the stale operation and offers the current result. If Pending Verification is already recorded, it shows that request without creating another transition. Otherwise the Technician reviews the current state before continuing.

Result: Latest accepted state. UI: UI-20E; UI-23A.

UC-04 — Verify Service Recovery

Primary actor: IT Administrator.

Supporting actors: Campus User, who indirectly provides the fresh connectivity measurements needed for verification.

Trigger: The IT Administrator opens an issue in the Pending Verification state from the issue queue to review its recovery status.

Preconditions: The IT Administrator has an authenticated, authorized session. An issue exists in the Pending Verification state following a recorded technician intervention. Fresh connectivity evidence from the affected zone may or may not be available.

Success postconditions: The issue is transitioned to Closed. The transition decision, closure reason, responsible actor and timestamp are recorded. The campus heatmap is updated to indicate verified service recovery for the affected zone.

Alternative postconditions: If the fault persists, the issue is Reopened with a recorded reason, retains its existing technician assignment, and appears in that technician's work queue. When the technician resumes work, the system records Reopened → In Progress. If the evidence is insufficient, the issue remains Pending Verification.

Main success flow

1. The IT Administrator requests the issue queue.

2. The system lists issues, including those currently in the Pending Verification state.

3. The IT Administrator selects a Pending Verification issue.

4. The system presents the issue identifier, affected zone, the technician's intervention record, and the aggregated fresh Wi-Fi measurement evidence gathered since the intervention.

5. The system presents the fresh measurement count, observation times and service-quality results, together with the configured evidence-sufficiency and service-recovery criteria.

6. The IT Administrator reviews the fresh evidence and authorizes closure only when its quantity, freshness, relevance and service-quality results meet the configured recovery criteria.

7. The system records the decision, closure reason, responsible administrator and timestamp. It transitions the issue state to Closed.

8. The system updates the coverage heatmap to reflect the verified service recovery in the corresponding zone.

Alternative and exception flows

A1 — Reopen unresolved issue: At Step 6, if the fresh measurements indicate that poor connectivity persists, the IT Administrator chooses to reopen the issue and supplies a non-empty reason. The system records the reason, actor and timestamp, transitions the issue to Reopened, retains its existing assignee, and makes it available in the technician's work queue. When the technician resumes work, the transition to In Progress is recorded.

A2 — Missing reopening reason: Following A1, if the IT Administrator attempts to reopen the issue without providing a reason, the system refuses to record the transition and requests a mandatory reason. The issue remains in Pending Verification.

A3 — Insufficient evidence: At step 5, if the system determines that the fresh measurements do not yet reach the required minimum count to verify recovery, it informs the IT Administrator. The administrator cannot authorize closure, and the system preserves the issue in the Pending Verification state while more evidence is sought.

Related functional requirements: FR-30 through FR-39.

UC-05 — Investigate Underobserved Zones

Primary actor: IT Administrator.

Supporting actor: AP Controller, an external read-only telemetry source. Campus users supply requested evidence through UC-01; their measurement submission is a separate goal.

Trigger: The IT Administrator requests to investigate a flagged zone or resumes an existing investigation after new evidence arrives.

Preconditions: The actor has an authenticated, authorized IT Administrator session. The zone and its stored investigation context can be read. A current AP Controller response is not required.

Success postconditions: A justified survey outcome and its actor, time and evidence references are recorded. A supported fault is linked to a new Detected candidate or a matching active issue. Adequately evidenced No fault found completes the investigation without creating an issue.

Other outcomes: An accepted targeted request may await responses. Insufficient evidence keeps the investigation open. Cancellation, denied access and failed recording preserve previously accepted records and WiFiIssue states.

Main success flow

1. The IT Administrator requests the flagged-zone list and selects a zone to investigate.

2. The system presents the flag reason, observation period, fresh evidence count, expected use, exclusions, mapped APs, related active issues and investigation progress.

3. The system attempts to retrieve relevant AP Controller telemetry and presents its status, timestamp and freshness or availability label. It excludes stale telemetry from current supporting evidence.

4. The IT Administrator specifies a target zone, collection period and instructions and requests targeted measurements.

5. The system validates and records the request, makes it available to the intended recipients, and confirms its identifier and delivery outcome. The investigation awaits evidence.

6. After evidence has been received through UC-01, the IT Administrator resumes the investigation and reviews the eligible measurements and their observation times.

7. The IT Administrator selects Fault supported and supplies a rationale and evidence references.

8. The system checks authorization, current investigation context, required information, evidence freshness and the configured fault criteria.

9. The system finds no matching active issue and records the survey outcome together with a link to a newly created candidate in Detected.

10. The system confirms the accepted outcome and candidate link. The IT Administrator may continue with UC-02; UC-05 ends without approving or assigning the candidate.

Alternative and exception flows

A1 No flagged zones — Step 1. If no zone currently meets the underobservation rule, the system presents an empty list with the evaluation time. The use case ends without creating an investigation or issue.

A2 Missing expected-use or mapping context — Step 2. The system identifies the missing profile, zone mapping or period context. The administrator may use the existing configuration functions and return, or record a deferral. An evidence-adequacy decision remains unresolved until the required context is available.

A3 Planned exclusion or expected low use — Step 2. The administrator checks the configured context. A confirmed planned exclusion can end the investigation with a reason while retaining the Planned exclusion label. Expected low use is considered in the evaluation; it does not by itself establish acceptable service. Otherwise resume Step 4.

A4 Stale or unavailable telemetry — Step 3. The system labels stale or unavailable telemetry and shows the last successful timestamp when available. Stale data is excluded from current supporting evidence. The administrator may continue at Step 4 using user evidence. A current offline report is supporting context, not proof of the physical cause or user-facing impact.

A5 Invalid targeted request — Steps 4–5. If the zone, period or instructions are missing or invalid, the system identifies the fields requiring correction. No request is accepted. The administrator corrects the information and resumes Step 4, or cancels.

A6 Targeted request cannot be recorded — Step 5. The system reports that the request was not saved and retains the entered values in the current form view. The administrator retries Step 4 or returns to the investigation. No accepted request or successful delivery is claimed.

A7 Request saved but delivery unconfirmed — Step 5. The accepted request remains available with its identifier and delivery limitation. The administrator may review or retry delivery of that request. Its saved record is preserved, and retry does not create a second measurement request.

A8 Responses remain insufficient — Steps 6 or 8. The system identifies the remaining evidence gap and retains the open investigation. The administrator can request further measurements, continue collecting, or defer review with a reason. The use case ends for this review without a no-fault conclusion or an automatic WiFiIssue transition.

A9 Adequate evidence supports no fault — Steps 7–9. The administrator selects No fault found and supplies a rationale. The system validates sufficient fresh evidence against the acceptable-service criteria, records the outcome, and completes the investigation without creating an issue. It does not close any existing WiFiIssue.

A10 A matching active issue exists — Step 9. The system links the accepted survey and relevant evidence to the matching issue instead of creating another candidate. The system confirms that link and preserves the issue's current state. Continue at Step 10.

A11 Outcome is invalid or cannot be recorded — Steps 8–9. Missing rationale, missing evidence references or an unsupported conclusion is identified for correction at Step 7. If storage fails, no accepted result is claimed; previously accepted data is preserved and the administrator can retry or review. The outcome and required issue link must both be accepted before success is shown.

A12 Access or investigation context changes — Before a protected read or write, the system checks the current role and investigation. Unauthorized operations are denied. A changed investigation is reloaded for review. An already accepted outcome is shown instead of being recorded again; no duplicate issue or second outcome is created.

A13 Administrator cancels — Before a request or outcome is accepted, the administrator cancels the current entry. The system returns to the investigation without accepting that entry. Earlier requests, evidence and outcomes remain available.

Related functional requirements: FR-40–49; supporting FR-01–09, FR-50, FR-53–54 and FR-56–59.

3.3.3 Object Model

Reserved for RAD-v2. The object model is intentionally not part of this RAD-v1 submission.

3.3.4 Dynamic Models

Reserved for RAD-v2. Sequence and state-machine diagrams are intentionally not part of this RAD-v1 submission.

3.3.5 User Interface Mock-ups

The working, navigable prototype uses smartphone frames (390 × 844) for Campus Users and Technicians and desktop frames (1440 × 900) for IT Administrators, as specified in NFR-24.

Figma design file and complete screen catalogue

Open the UC-05 interactive prototype

Access settings checked on 25 October 2026: Anyone can view. The supplied share link is provided above. Representative screens are embedded below and in Annex III so the report can also be read independently of Figma.

The design specification calls for a shared component library (buttons, status chips, telemetry badges, banners, menus, form fields, heatmap cells, and issue and notification cards) with common color and text styles. The shared components are intended to meet WCAG 2.1 AA contrast ratios (NFR-02), and every WiFiIssue state and heatmap category must have a text label in addition to color (NFR-03). Screen IDs follow these ranges: UI-01–UI-09 for UC-01; UI-10–UI-19 for UC-02; UI-20–UI-29 for UC-03; UI-30–UI-39 for UC-04; UI-40–UI-49 for UC-05; and UI-50 and above for shared/supporting screens. Letter suffixes identify variants (for example, UI-12B). These IDs are used in Section 3.4.

Table 1 lists the prototype flows, the screens on their main path, and the error and recovery states they show.

Table 1. Prototype flows, main paths, and error and recovery states.

Flow	Actor(s)	Main path (screens)	Error and recovery states shown

Flow	Actor(s)	Main path (screens)	Error and recovery states shown
Campus user: sign-in, consent, home	Campus user	UI-50 Sign In → UI-51 Privacy Notice and Consent → UI-52 Home; UI-59 Create Account; UI-62 Profile and Privacy Settings	UI-62B Consent Withdrawn
UC-01 Submit Connectivity Evidence	Campus user	UI-52 → UI-01 Select Zone and Start Test → UI-02 Test in Progress → UI-03 Test Result and Submit → UI-05 Submission Confirmation; UI-07 My Submissions	UI-04 Report Failed Connection; UI-06A/UI-06B Offline Queue
UC-02 Triage and Assign an Issue	IT administrator	UI-53 Dashboard → UI-10 Issue Queue → UI-11 Issue Details and Evidence → UI-12 Decision and Assignment → UI-12G Assign Technician → UI-13 Assignment Result	Rejection with reason; Approved · Unassigned (no eligible technician); assignment not recorded; notification failed; stale or unavailable telemetry; UI-19 Access Denied
UC-03 Record an Intervention	Technician	UI-54 My Assigned Issues → UI-20 Assigned Issue Detail → UI-21 Start Work → UI-21A Work in Progress → UI-22 Record Diagnosis and Actions → UI-22A Completed Intervention Form → UI-23 Request Verification → UI-23A Verification Requested → UI-24 My Issues After Verification	UI-22B Required Field Missing; UI-21B Start Work Save Failed; UI-22C Intervention Save Failed; UI-23B Verification Request Failed; UI-20B Stale AP Telemetry; UI-20C Access Denied; UI-20E Issue State Changed; UI-20D Saved Intervention History
UC-04 Verify Service Recovery	Campus user, IT administrator	Campus-user evidence request: UI-58 → UI-30 Verification Request (test CTA displayed). IT review: UI-53 → UI-31 Verification Review → UI-32 Close Issue or UI-33 Reopen Issue with Reason.	UI-34 Pending – Insufficient Evidence is a separate snapshot. In the inspected prototype, UI-30 test start, UI-33 submit, and UI-32/UI-34 return CTAs are not linked; UI-33 Cancel returns to UI-31.
UC-05 Investigate Underobserved Zones	IT administrator	UI-53 Dashboard → UI-40 Underobserved Zones List → UI-41 Zone Investigation Detail → UI-42 Request Targeted Measurements → UI-42A Targeted Request Recorded → UI-43 Record Survey Outcome → UI-43A Fault Supported	UI-40A empty list; UI-41A stale telemetry; UI-41B unavailable telemetry; UI-41C exclusion/missing context; UI-42B request not saved; UI-42C delivery unconfirmed; UI-42D invalid request; UI-43B no fault found; UI-43C insufficient evidence; UI-43D invalid outcome; UI-43E outcome not saved; UI-49 access/context changed
Coverage heatmap (supporting)	Campus user, technician, IT administrator	UI-55 Coverage → UI-57 Zone Detail (campus user); UI-55T Coverage (technician); UI-56 Coverage with zone panel (IT administrator)	UI-56B AP Controller Unreachable
Technician home and notifications	Technician	UI-54 My Assigned Issues; UI-55T Coverage; UI-63 Notifications	—
Configuration (supporting)	IT administrator	UI-60 Thresholds and Freshness ↔ UI-61 AP-to-Zone Mapping and Expected Use	—

The coverage heatmap supports all three roles. Zones use color, category labels and sample counts (NFR-03). Too few recent measurements means “insufficient evidence”, never good coverage; few complaints do not prove service quality. Planned exclusions are hatched. Selecting a zone shows measurements, expected use, AP status and active issues. If telemetry is stale or unavailable, a banner gives its last update; the map continues using user measurements (NFR-08, NFR-21).

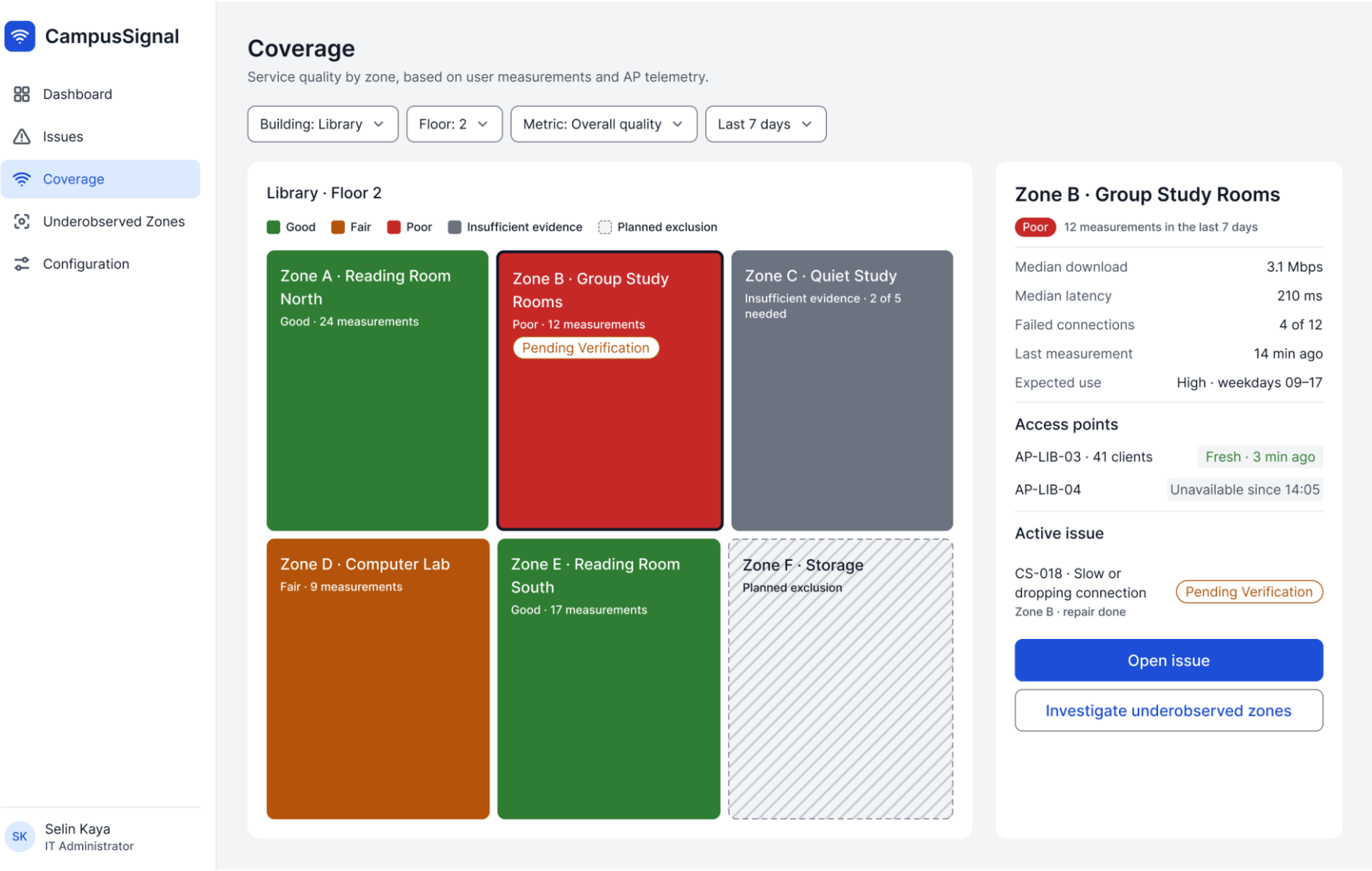


Figure 2. UI-56 Coverage Heatmap for IT administrators. Library, Floor 2; the map distinguishes five quality categories and the selected Zone B panel shows measurements, telemetry freshness and issue CS-018.

Privacy requirements are visible in the prototype: a campus user gives consent on UI-51 before the first measurement, can withdraw it on UI-62 (UI-62B shows the resulting state), and the location of a measurement is always selected at zone level (NFR-17, NFR-18). IT administrators change detection thresholds, the telemetry freshness threshold, AP-to-zone mappings, and expected-use profiles on UI-60 and UI-61 without software changes (NFR-23).

The following specifications define the UC-02 screen contents. The screen IDs are stable traceability identifiers. These specifications describe the interface and do not imply that a backend, live AP Controller connection, or notification service has been implemented.

The Figma prototype demonstrates screen states and navigation. Its populated fields, receipts and error views are illustrative snapshots; they do not establish backend storage, live AP telemetry, notification delivery, periodic evaluation, timing, data-transfer limits or retention enforcement. These remain implementation acceptance targets. The UC-01 snapshots illustrate the existing-issue matching alternative, while the UC-01 textual main flow specifies creation of a new Detected candidate.

UI-10 — Issue Queue

Purpose: let the IT Administrator identify work awaiting triage and approved issues awaiting assignment. Display issue ID, affected zone, submission/observation time, current state, and assignee if any. Provide a way to open issue details and distinguish Detected issues from Approved-but-unassigned issues.

UI-11 — Issue Details and Evidence

Purpose: support a reasoned decision. Display connectivity reports and measurements, timestamps, affected zone, mapped AP, and available AP Controller information. Show AP-reported Online/Offline separately from telemetry Stale/Unavailable. When telemetry is missing, explain that the cause is unknown; do not label the AP offline solely because data is absent.

UI-12 — Decision and Assignment

Purpose: record an approval rationale or a mandatory rejection reason for a Detected issue. In the main prototype path, Approve & continue to assignment opens UI-12G, where an eligible Technician and assignment note are presented. UI-12H illustrates the alternative Technician selection. Assignment is permitted only for an Approved issue without a current assignee. An assignment note may request physical inspection of the AP, power or local connectivity.

UI-13 — Assignment Result

Purpose: make the outcome unambiguous. Show the resulting issue state, assigned technician if any, assignment note, recorded decision/assignment information, and notification result. If no technician is available, show Approved / Unassigned. If notification delivery fails after assignment is stored, retain Assigned and show the notification issue.

UC-02 navigation: UI-10 Issue Queue → UI-11 Issue Details and Evidence → UI-12 Decision and Assignment → UI-12G Assign Technician → UI-13 Assignment Result. Rejection branches through UI-12B to a Rejected result. Leaving an Approved issue unassigned produces an Approved / Unassigned result; the issue can be opened again from the queue for a later assignment attempt.

UC-05 screen specification

The UC-05 screen specification uses 1440 × 900 desktop frames and instances from 01 Components. The shared sidebar, navigation and main components remain common. The main path is UI-40 → UI-

41 → UI-42 → UI-42A → UI-43 → UI-43A; No Fault Found and Insufficient Evidence branch to UI-43B and UI-43C.

Investigation progress is distinct from WiFiIssue state. A supported new candidate is Detected and enters UC-02; an adequately evidenced no-fault result completes the investigation without closing an existing issue. Missing or stale telemetry never establishes good service or a fault by itself.

Screen specification	Required content
UI-40 Underobserved Zones List	Zone entries, observation period, fresh count, expected-use context, flag reason and progress. Open a zone.
UI-40A No Underobserved Zones	Empty result with last evaluation time and a return route.
UI-41 Zone Investigation Detail	Expected use, exclusions, measurements, active issues and AP context. Review telemetry and request measurements or record an outcome.
UI-41A Stale Telemetry	Stale Telemetry Badge and Telemetry Stale Banner with last update and age. Continue using eligible user evidence.
UI-41B Telemetry Unavailable	Unavailable Telemetry Badge and AP Unreachable Banner. AP status is unknown; last successful timestamp if known.
UI-41C Investigation Context	Separate labelled examples for a planned exclusion and missing profile or mapping context. Record the reason or defer; do not infer good service.
UI-42 Request Targeted Measurements	Zone, collection period and instructions in Text Field instances. Submit or cancel.
UI-42A Targeted Request Recorded	Saved request identifier, time, delivery outcome, response count and Awaiting measurements progress.
UI-42B Targeted Request Not Saved	Entered values retained for this form; retry recording or return.
UI-42C Request Delivery Unconfirmed	Saved request identifier retained. Retry delivery or review the accepted request.
UI-42D Required Request Information Missing	Text Field Error variants for invalid period or missing instructions. Correct and resubmit or cancel.
UI-43 Record Survey Outcome	Fresh evidence and configured criteria, selected outcome, rationale and evidence references. Save or cancel.
UI-43A Fault Supported	Accepted survey identifier and link to a new Detected candidate, or to the matching active issue with its existing state.
UI-43B No Fault Found Investigation Complete	Adequate fresh evidence, saved rationale and completed investigation. No new issue; no existing issue closed.
UI-43C Insufficient Evidence	Remaining gap and open investigation. Continue collecting, request more measurements or defer with a reason.
UI-43D Required Outcome Information Missing	Text Field Error variants and correction route. No accepted outcome.
UI-43E Survey Outcome Not Saved	No success receipt; retain current form values and earlier records. Retry or review.

Screen specification	Required content
UI-49 Access Denied or Investigation Changed	Explain denied access or changed context. Review an already accepted result or return safely without a duplicate write.

Cancel returns to the selected investigation. A failed request save retries recording, while a delivery failure retries delivery of the existing request. Failed outcome recording preserves earlier records and supplies a retry route.

3.4 Traceability and Consistency

Requirement ID	Source / Stakeholder	Scenario / Use Case	Analysis and Dynamic Model Elements	UI Screen / Status
FR-01	Campus User	SC-01 / UC-01	Submit connectivity measurement or failure report	UI-01, UI-04
FR-02	Campus User	SC-01 / UC-01	Capture location (building, floor, zone) and evidence	UI-01, UI-03
FR-03	IT Administrator	SC-01 / UC-01	Evaluate evidence against thresholds and create issue	Background process / Candidate issue creation
FR-04	IT Administrator	SC-01 / UC-01	Append evidence to existing active issue	Background process / Issue matching
FR-05	IT Administrator	SC-01 / UC-01	Retain evidence without creating an issue (non-fault)	Background process / Heatmap log
FR-06	Campus User	SC-01 / UC-01	Provide submission feedback / confirmation	UI-05
FR-07	Campus User	SC-01 / UC-01	Offline queue detection (local save)	UI-06A
FR-08	Campus User	SC-01 / UC-01	Offline queue notification	UI-06A
FR-09	Campus User	SC-01 / UC-01	Background synchronization on reconnection	UI-06B

Requirement ID	Source / Stakeholder	Scenario / Use Case	Analysis and Dynamic Model Elements	UI Screen / Status
FR-10	IT Administrator	SC-02 / UC-02	WiFiIssue queue; detailed classes/sequence model in RAD-v2	UI-10; Detected and Approved / Unassigned
FR-11	IT Administrator	SC-02 / UC-02	WiFiIssue evidence; model details in RAD-v2	UI-11; evidence and timestamps
FR-12	IT Administrator; AP Controller	SC-02 / UC-02	AP telemetry retrieval attempt and result classification; included in UC-02; dynamic-model detail in RAD-v2	UI-11; Online/Offline vs Stale/Unavailable
FR-13	IT Administrator	SC-02 / UC-02	WiFiIssue decision history; state Detected → Approved	UI-12; approval rationale
FR-14	IT Administrator	SC-02 / UC-02	WiFiIssue rejection history; state Detected → Rejected	UI-12 and UI-13; rejection reason
FR-15	IT Administrator	SC-02 / UC-02	Assignment eligibility rule; detailed model in RAD-v2	UI-12G/UI-12H and UI-12E/UI-12F assignment views; Approved and unassigned only
FR-16	IT Administrator	SC-02 / UC-02	Technician assignment; inspection task note	UI-12G/UI-12H technician selection and assignment note; UI-12E/UI-12F retry selection
FR-17	IT Administrator	SC-02 / UC-02	WiFiIssue assignment history; state Approved → Assigned	UI-12G/UI-12H → UI-13/UI-13F; recorded assignee and assignment note
FR-18	Technician; IT Administrator	SC-02 / UC-02	Assignment notification outcome; detailed model in RAD-v2	UI-13; notification result
FR-19	IT Administrator	SC-02 / UC-02	WiFiIssue remains Approved without an assignee	UI-10 and UI-13; Approved / Unassigned
FR-20	Technician; IT where applicable	SC-03 / UC-03	Assigned WiFiIssue and Technician queue; detailed analysis/dynamic models in RAD-v2	UI-54; UI-24; UI-24A; UI-24B
FR-21	Technician; IT where applicable	SC-03 / UC-03	WiFiIssue, connectivity evidence, AP telemetry and last-update label; model detail in RAD-v2	UI-20; UI-20B
FR-22	Technician; IT where applicable	SC-03 / UC-03	Work-start event and transition Assigned/Reopened → In Progress; model detail in RAD-v2	UI-21; UI-21A; UI-21B
FR-23	Technician; IT where applicable	SC-03 / UC-03	Intervention record: diagnosis, actions taken, actor and timestamp; model detail in RAD-v2	UI-22; UI-22A
FR-24	Technician; IT where applicable	SC-03 / UC-03	Required-field validation for diagnosis and actions taken; model detail in RAD-v2	UI-22B
FR-25	Technician; IT where applicable	SC-03 / UC-03	Read-only Intervention history linked to WiFiIssue, Technician and recording time; corrections recorded as new entries; detailed models in RAD-v2	UI-20D Saved Intervention History; UI-24B My Issues After Intervention
FR-26	Technician; IT where applicable	SC-03 / UC-03	VerificationRequest linked to the assigned In Progress WiFiIssue and its accepted intervention; detailed models in RAD-v2	UI-23 Request Verification; UI-23A Verification Requested
FR-27	Technician; IT where applicable	SC-03 / UC-03	Transition In Progress → Pending Verification and hand-off to UC-04; model detail in RAD-v2	UI-23A; UI-24

Requirement ID	Source / Stakeholder	Scenario / Use Case	Analysis and Dynamic Model Elements	UI Screen / Status
FR-28	Technician; IT where applicable	SC-03 / UC-03	Issue assignment/state and duplicate-start protection; model detail in RAD-v2	UI-20C; UI-20E
FR-29	Technician; IT where applicable	SC-03 / UC-03	Accepted-record preservation and retry/review outcomes; model detail in RAD-v2	UI-21B; UI-22C; UI-23B; UI-20E
FR-30	IT Administrator	SC-04 / UC-04	Display post-intervention user measurements for verification	UI-30, UI-31
FR-31	IT Administrator	SC-04 / UC-04	Evaluate fresh evidence against acceptance criteria	UI-31
FR-32	IT Administrator	SC-04 / UC-04	WiFiIssue closure; state Pending Verification → Closed	UI-31, UI-32
FR-33	IT Administrator	SC-04 / UC-04	Retain issue when evidence is insufficient; state Pending Verification	UI-31, UI-34
FR-34	IT Administrator	SC-04 / UC-04	Reopen unresolved issue; state Pending Verification → Reopened	UI-31, UI-33
FR-35	IT Administrator	SC-04 / UC-04	Mandatory reopening reason validation	UI-33
FR-36	IT Administrator; Technician	SC-04 / UC-04	Retain technician assignment after reopening; Reopened → In Progress when work resumes	UI-33 reopening form; FR-36 and UC-03 specify retaining the assignee and resuming Reopened → In Progress. The current prototype has no linked Reopen submission or dedicated resumption snapshot.
FR-37	IT Administrator	SC-04 / UC-04	Record verification transition history (actor, timestamp, reason)	UI-32, UI-33
FR-38	IT Administrator; All Users	SC-04 / UC-04	Update coverage heatmap upon closure	UI-32, UI-55, UI-56
FR-39	IT Administrator	SC-04 / UC-04	Preserve Pending Verification state across sessions	UI-31, UI-34
FR-40	IT Administrator	SC-05 / UC-05	Underobservation evaluation and investigation context (analysis models deferred to RAD-v2)	UI-40
FR-41	IT Administrator	SC-05 / UC-05	Flagged-zone list and investigation progress (analysis models deferred to RAD-v2)	UI-40, UI-40A
FR-42	IT Administrator	SC-05 / UC-05	Zone context, expected use, exclusions and evidence (analysis models deferred to RAD-v2)	UI-41, UI-41A, UI-41B, UI-41C
FR-43	IT Administrator; AP Controller	SC-05 / UC-05; SC-02 / UC-02	Read-only telemetry and freshness classification (analysis models deferred to RAD-v2)	UI-41A, UI-41B, UI-41C, UI-11, UI-56B
FR-44	IT Administrator; Campus User	SC-05 / UC-05; SC-01 / UC-01	Targeted measurement request and collection period (analysis models deferred to RAD-v2)	UI-42, UI-42D
FR-45	IT Administrator; Campus User	SC-05 / UC-05; SC-01 / UC-01	Request delivery, response association and collection progress (analysis models deferred to RAD-v2)	UI-42A, UI-42B, UI-42C

Requirement ID	Source / Stakeholder	Scenario / Use Case	Analysis and Dynamic Model Elements	UI Screen / Status
FR-46	IT Administrator	SC-05 / UC-05	Record justified survey outcome and evidence references (analysis models deferred to RAD-v2)	UI-43, UI-43A, UI-43C
FR-47	IT Administrator	SC-05 / UC-05 → UC-02	Link supported fault to candidate issue; UC-05 does not approve or assign it	UI-43A
FR-48	IT Administrator	SC-05 / UC-05	Complete adequately evidenced no-fault investigation without creating a WiFiIssue	UI-43B
FR-49	IT Administrator	SC-05 / UC-05	Retain investigation with unresolved evidence/configuration gap	UI-43C
FR-50	University IT	All	Secure authentication (NFR-13/14 compliance)	UI-50 (Shared sign-in)
FR-51	IT Administrator	All	Heatmap visualization	UI-55 / UI-56 (Coverage heatmap)
FR-52	All Users	All	Heatmap status differentiation (text/icons + color)	UI-55 / UI-56
FR-53	IT Administrator	All	Access point mapping (AP to zone)	UI-61 (Configuration)
FR-54	IT Administrator	All	Expected-use profiling per zone	UI-61 (Configuration)
FR-55	Campus User	All	Self-registration (Campus User role restriction)	UI-59 (Account creation)
FR-56	Campus User	UC-01	Informed consent collection & withdrawal	UI-51 / UI-62 (Consent & Settings)
FR-57	Campus User	UC-01	View own submissions and issue status history	UI-07 My Submissions
FR-58	IT Administrator	All	Configure detection thresholds, windows, and rules	UI-60 (Configuration)
FR-59	AP Controller	All	Retrieve AP telemetry via read-only interface	Telemetry integration / UI-11
NFR-01	Campus user	UC-01, UC-04	Evidence-submission interaction budget; acceptance target to be measured on the complete home-to-confirmation route	UI-52 → UI-01 → UI-02 → UI-03 → UI-05; test progression is specified as automatic, and the home/test-start/submit actions illustrate the three-interaction budget. Completion/timing is not certified by this report.
NFR-02	All users; WCAG 2.1	All	—	Shared color styles and labelled component variants; full-screen WCAG AA contrast compliance remains an acceptance target
NFR-03	All users; WCAG 2.1	UC-02, UC-05; heatmap	Zone quality category; WiFiIssue state	UI-55, UI-55T, UI-56, UI-57; status chips with labels
NFR-04	All users	UC-01, UC-02, UC-04	—	Error and recovery screens, e.g. UI-19, UI-56B, UI-62B

Requirement ID	Source / Stakeholder	Scenario / Use Case	Analysis and Dynamic Model Elements	UI Screen / Status
NFR-05	Campus user; IT administrator	All	—	— (acceptance target)
NFR-06	IT administrator; technician	UC-01–UC-04	Evidence and WiFiIssue records	— (acceptance target)
NFR-07	Campus user; proposal (offline queue)	UC-01	Offline evidence queue	UI-06A queued evidence; UI-06B synchronized evidence; UI-62/UI-62B consent withdrawal. The 24-hour persistence and one-minute retry limits require implementation testing.
NFR-08	IT administrator; AP Controller (A-01)	UC-01, UC-02, UC-04, UC-05	AP telemetry status and last update	UI-11 (telemetry unavailable), UI-56B (AP unreachable banner)
NFR-09	Campus user	UC-01, UC-04	Connectivity test	UI-02 Test in Progress; 20-second and 10-MB limits remain implementation acceptance targets
NFR-10	Campus user; IT administrator	UC-01; heatmap	—	— (acceptance target)
NFR-11	IT administrator; proposal	UC-01	Detection of candidate issues	— (acceptance target)
NFR-12	IT administrator; proposal	UC-05	Underobserved-zone evaluation	UI-40 evaluated-zone list and evaluation timestamp; at-least-daily execution remains an implementation acceptance target
NFR-13	University IT	All	User account; roles	UI-50, UI-50B, UI-59
NFR-14	University IT	All	Role permissions	Role home screens UI-52, UI-53, UI-54; UI-19 Access Denied
NFR-15	University IT	All	—	— (interface constraint)
NFR-16	IT administrator; proposal	UC-02, UC-03, UC-04	WiFiIssue transition history	UI-13 (issue status history)
NFR-17	Campus user; KVKK	UC-01	Consent record	UI-51, UI-62, UI-62B
NFR-18	Campus user; proposal (zone-level location)	UC-01	Measurement location (zone level)	UI-01 zone selectors and UI-03 result location; excluded data categories require record/schema inspection
NFR-19	Campus user; KVKK	UC-01, UC-02, UC-03	Submitter link and retention	UI-11 IT evidence view; UI-20 Technician evidence view with identities hidden. The 180-day submitter-link removal requires implementation/data inspection.
NFR-20	University IT; proposal (read-only API)	UC-01, UC-02, UC-04, UC-05	AP Controller interface	— (interface constraint)
NFR-21	IT administrator; proposal	UC-01, UC-02, UC-04, UC-05	Telemetry freshness rule	Telemetry badges on UI-10, UI-11, UI-53, UI-55T, UI-56, UI-56B; setting on UI-60

Requirement ID	Source / Stakeholder	Scenario / Use Case	Analysis and Dynamic Model Elements	UI Screen / Status
NFR-22	IT administrator; proposal	UC-01, UC-04, UC-05	Test endpoint	— (interface constraint)
NFR-23	IT administrator	UC-01, UC-04, UC-05	Configuration	UI-60, UI-61
NFR-24	All users	All	—	Mobile and desktop frames in the prototype

The RAD-v1 traceability mapping covers FR-01–FR-59 and NFR-01–NFR-24. Analysis-class and dynamic-model references are addressed in RAD-v2. Screen identifiers are used consistently across the UI specifications and traceability mapping.

4 Glossary

Term	Meaning
Access Point AP	A network device that provides wireless access in a mapped campus area.
AP Controller	The external system supplying per-AP reported status, aggregate connected-client counts and timestamps through a read-only interface.
AP to zone mapping	The maintained association between an access point and one or more campus zones used to relate telemetry to reported locations.
Campus User	A student or staff member who submits connectivity evidence, explores coverage or responds to measurement requests.
IT Administrator	The authorized role that triages and assigns issues, reviews verification, closes or reopens issues and maintains configuration.
Technician	The authorized role that views and updates issues assigned to them and records diagnoses, interventions and verification requests.
Zone	A defined campus area identified at building, floor and zone level for measurements, expected use and coverage analysis.
WiFiIssue	The persistent record of a suspected or accepted Wi-Fi problem, its evidence, assignment, intervention history and current workflow state.
Connectivity evidence	A recorded user measurement or explicit failed-connection report, including its zone and observation time.
Connectivity measurement	A test observation of connectivity quality collected against the controlled campus test endpoint.
Failed connection report	Explicit evidence that a user could not complete a connection or connectivity test in the selected zone.
Observation time	The time the connectivity event or measurement occurred; it remains distinct from the later receipt time of an offline report.
Telemetry	The AP status, aggregate connected-client count and associated timestamp supplied by the AP Controller.
Fresh telemetry	Telemetry whose controller timestamp is within the configured freshness threshold in NFR-21.
Stale telemetry	Telemetry older than the configured NFR-21 threshold. The current default is more than 15 minutes; stale readings are labelled and excluded from current detection, verification and underobservation evidence.
Unavailable telemetry	Telemetry that cannot currently be obtained. It does not establish that the access point itself is offline.
Reported offline AP	An AP explicitly reported Offline by the AP Controller, with a timestamp. This is distinct from missing or stale telemetry and does not identify the physical cause.
Aggregate connected-client count	The controller-reported total of connected clients for an AP. This count alone establishes neither actual occupancy nor a connectivity fault.
Fresh measurement	A user measurement accepted as recent enough for its assessment purpose. Verification requires measurements relevant to the intervention and its follow-up; UC-04 defines the measurement window and acceptance rules. The telemetry threshold is not automatically the measurement threshold.

Term	Meaning
Insufficient evidence	Evidence that does not meet the configured quantity, freshness or relevance needed for a decision. It does not establish either adequate service or a fault.
Underobserved zone	A zone flagged because the available observations are insufficient in relation to expected use and intended service availability; investigation is required before concluding that a fault exists.
Expected use profile	The information describing expected use of a zone, used with observation counts and service expectations to interpret low activity.
Planned exclusion	A zone or area deliberately excluded from the intended service coverage, shown separately from poor service or insufficient observations.
Heatmap	The coverage view showing quality categories, evidence counts and freshness, with distinct treatment of insufficient evidence and planned exclusions.
Intervention	An accepted Technician record of diagnosis and actions taken for a specific issue.
Diagnosis	The Technician's recorded finding or explanation of the observed cause. It may identify an unresolved cause rather than assert a successful repair.
Actions taken	The recorded work performed by the Technician. CampusSignal records this account and does not itself perform physical repairs or AP reconfiguration.
Verification request	A recorded request to assess service recovery after an intervention, linked to its issue and intervention.
Service recovery verification	The UC-04 assessment of fresh evidence against configured criteria, supporting IT closure, reopening or continued waiting.
Transition history	The append-only history of WiFiIssue state changes, identifying the responsible actor, time and reason.
Append-only record	A retained entry that cannot be edited or deleted. Corrections are represented by additional entries, as required by NFR-16.
Offline queue	The campus-user mechanism that retains unsent connectivity evidence locally and transmits it after connectivity returns. This does not imply an offline technician-editing feature.
Internal scheduler	A system-internal mechanism that initiates periodic evaluations. Its work is described as system behavior; it is not an external actor.
Read-only controller interface	An external interaction limited to retrieving permitted AP information. It issues no restart or configuration commands to the controller.
Detected	A candidate issue has been created for IT review.
Approved	IT has accepted the issue; assignment has not yet been recorded. An issue may remain here when no eligible Technician is available.
Assigned	An issue has a recorded Technician assignment and work has not yet been recorded as started.
In Progress	The assigned Technician has started work. Recording an intervention alone leaves the issue in this state.
Pending Verification	A verification request has been recorded and the issue awaits sufficient evidence and IT review.
Closed	The IT Administrator has authorized closure after sufficient fresh evidence satisfies the configured acceptance criteria.
Rejected	IT has rejected a Detected candidate issue and recorded a reason.
Reopened	Verification showed that the fault persists. The issue retains its assigned Technician and appears in the Technician’s work queue; when work resumes, it transitions to In Progress.

5 Appendix

Annex I – Distribution of Work

This summary presents each member's primary ownership and collaborative responsibilities for the RAD-v1 deliverables.

Member	Responsibilities and documented contributions
Eren Öztürk	Prepared the Introduction and contributed the UC-01 scenario and textual use-case flow. Owned FR-01–FR-09 and supporting FR-50–FR-59, including connectivity evidence, submission feedback, offline queuing and synchronization. Coordinated the UC-01 screen requirements and requested integration support where needed.
Efe Özdemir	Prepared the Current System section, including the current eduroam support path, involved roles and pain points. Owned the UC-02 scenario and description, FR-10–FR-19, and the top-level use-case diagram. Maintained the editable PlantUML source and reviewed consistency between the UC-02 workflow, status terms and screen identifiers UI-10–UI-13 / UI-19.
Arda Korkmaz	Owned NFR-01–NFR-24 and the shared Figma structure, component library and navigation conventions. Coordinated the visual integration of role screens, common status and telemetry components, coverage-heatmap presentation, report assembly and final prototype checks. Added the final UI-56 export and share link to the integrated report. Prototype verification limits are recorded in Annex III.
Azize Görür	Owned the UC-04 scenario and description, FR-30–FR-39 and the associated verification screens. Consolidated the traceability mapping and reviewed state consistency across verification, closure, reopening and pending outcomes. Worked to keep the same issue-state terminology across requirements, use cases and screen paths.
Caner Gündaş	Owned the UC-05 scenario and description, FR-40–FR-49, and the underobserved-zone investigation screen specification. Consolidated the functional-requirement register and checked that targeted measurement requests, investigation outcomes and exception paths were represented. Compiled the distribution-of-work and meeting-record material for the RAD.
Devran Deniz Akgül	Owned the UC-03 scenario and description, FR-20–FR-29 and the glossary. Documented the technician's diagnosis/action recording and request-verification flow, contributed to the UI-20–UI-29 screen set, and reviewed navigation terminology. A Coverage → My Issues navigation fix was reported on 25 October and should be confirmed in the integrated prototype.

Annex III Representative Prototype Screens

The following native Figma exports illustrate the report's screen specifications. They show UI states; backend execution and unverified transitions remain acceptance work.

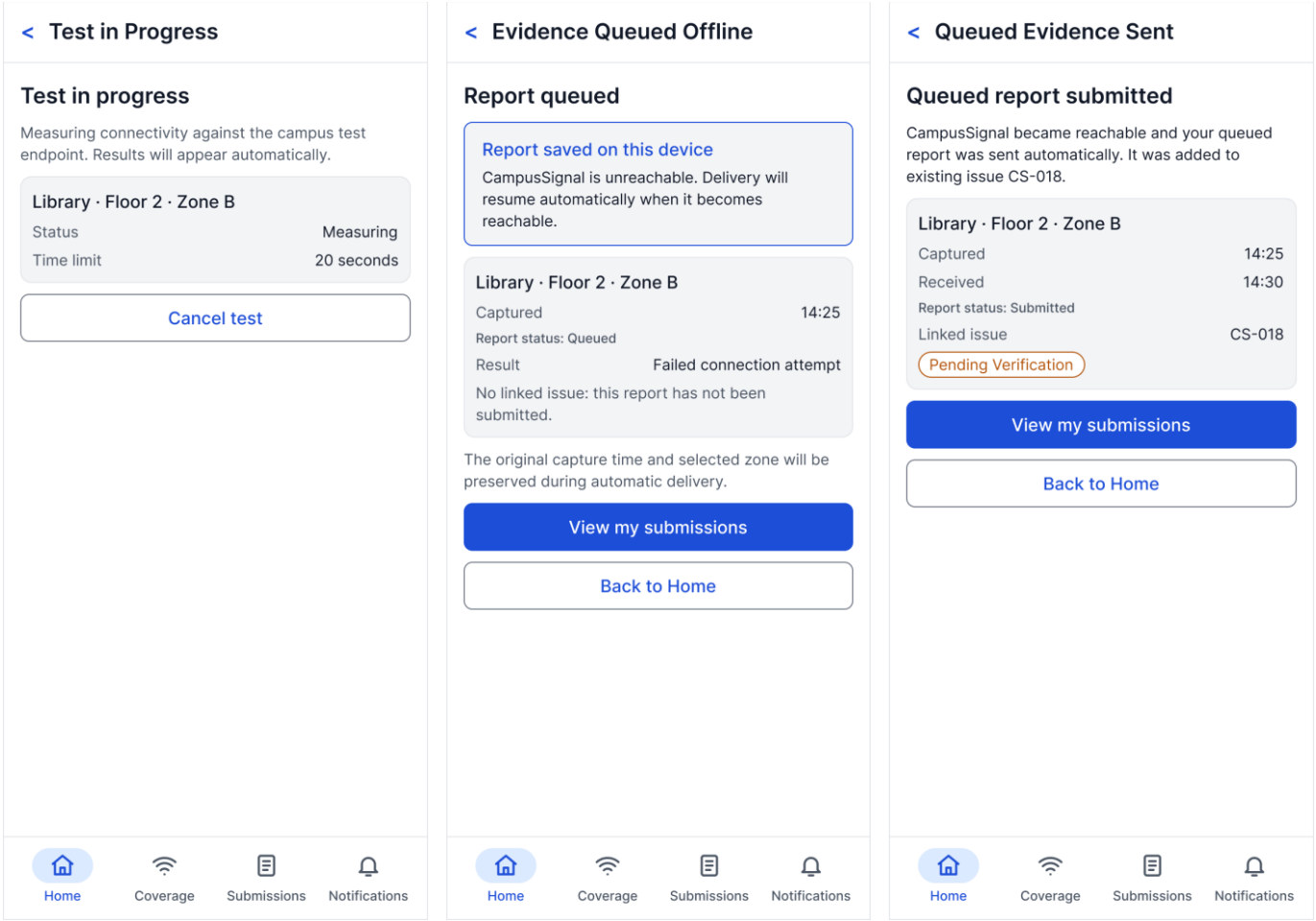


Figure 3. UC-01: UI-02 test in progress, UI-06A evidence queued offline and UI-06B queued evidence sent. Persistence and synchronization timing require implementation tests.

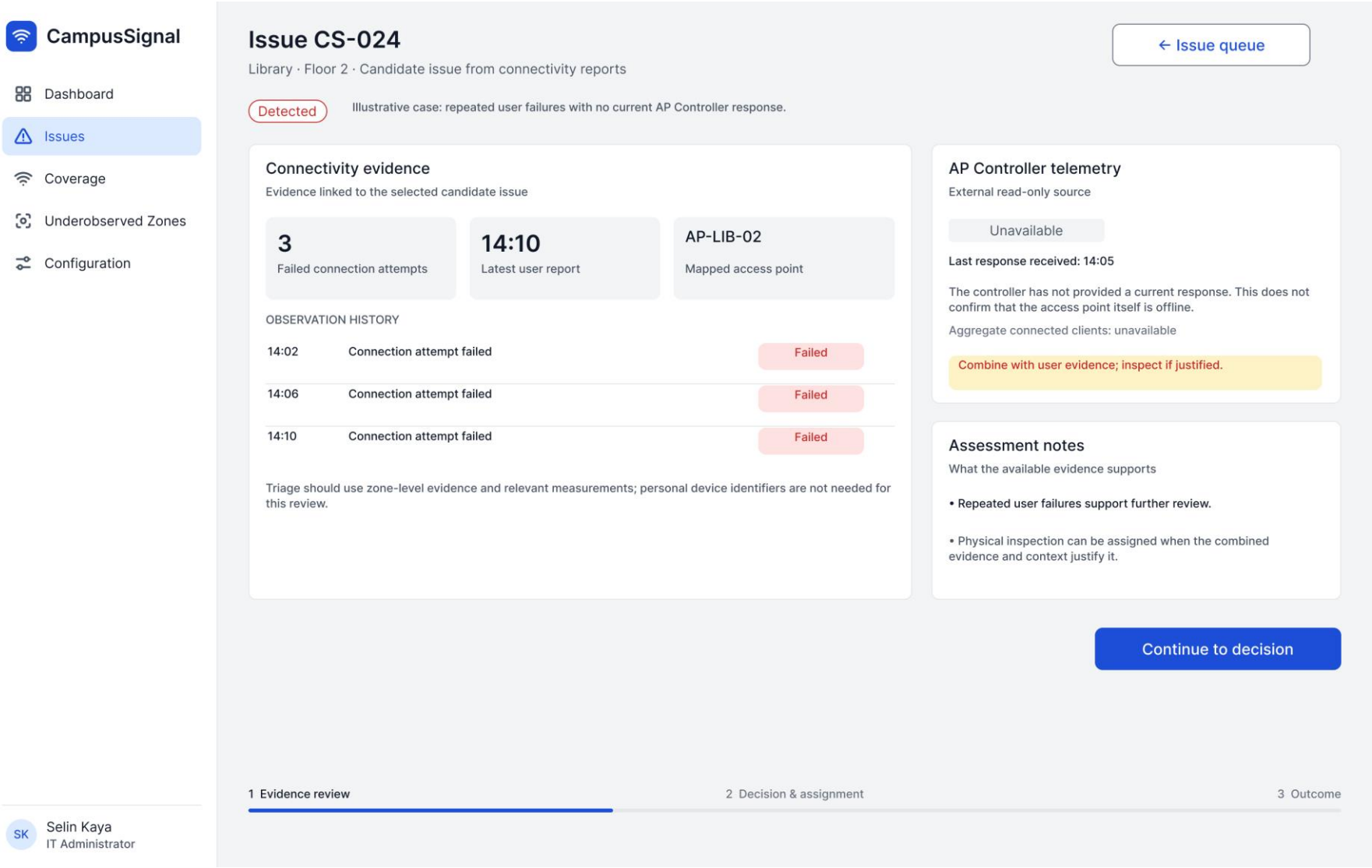


Figure 4. UC-02: UI-11 Issue Details and Evidence. Approval continues through UI-12 and UI-12G before UI-13 Assignment Result.

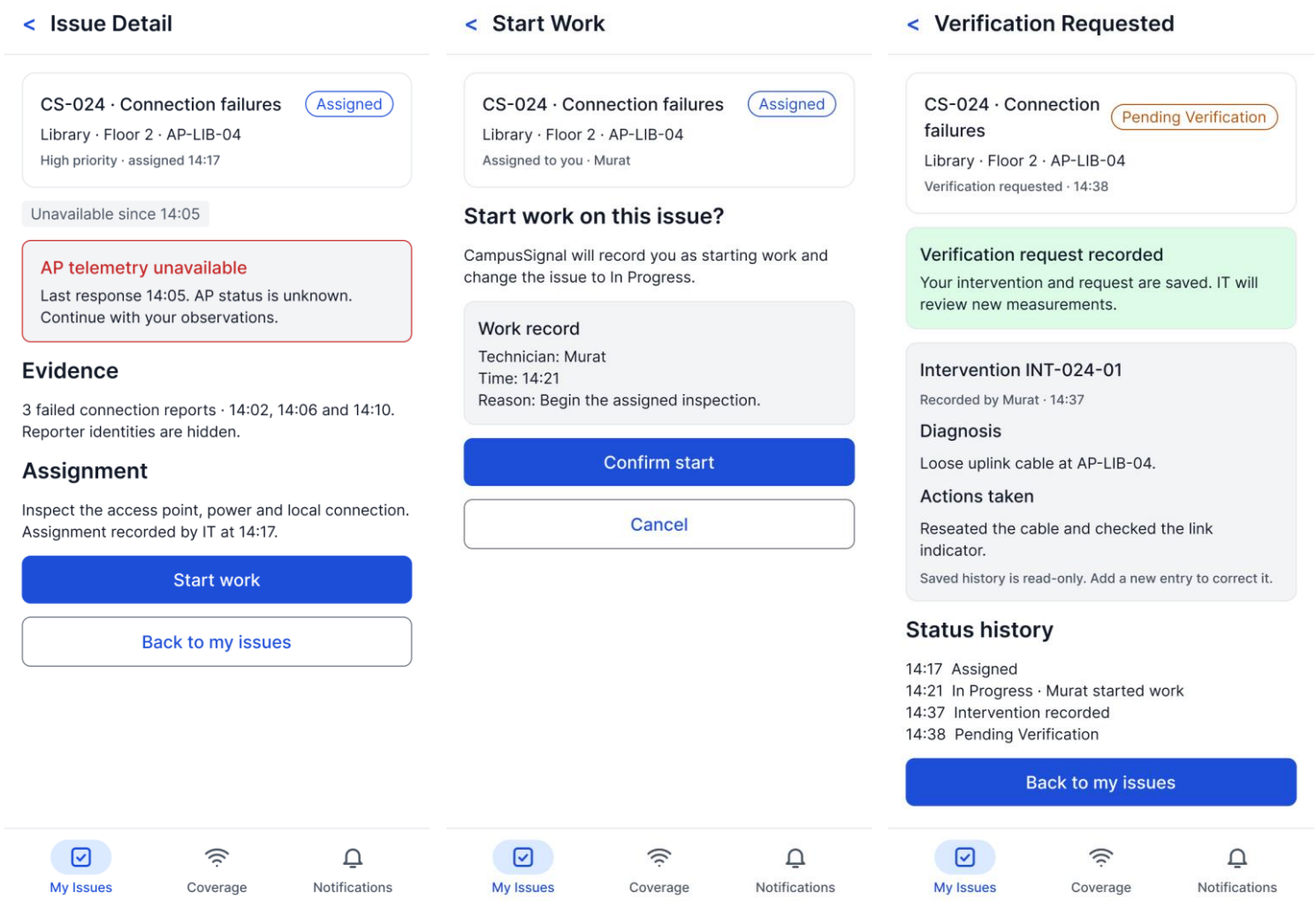


Figure 5. UC-03: UI-20 Issue Detail, UI-21 Start Work and UI-23A Verification Requested. The assigned technician records work and requests IT verification.

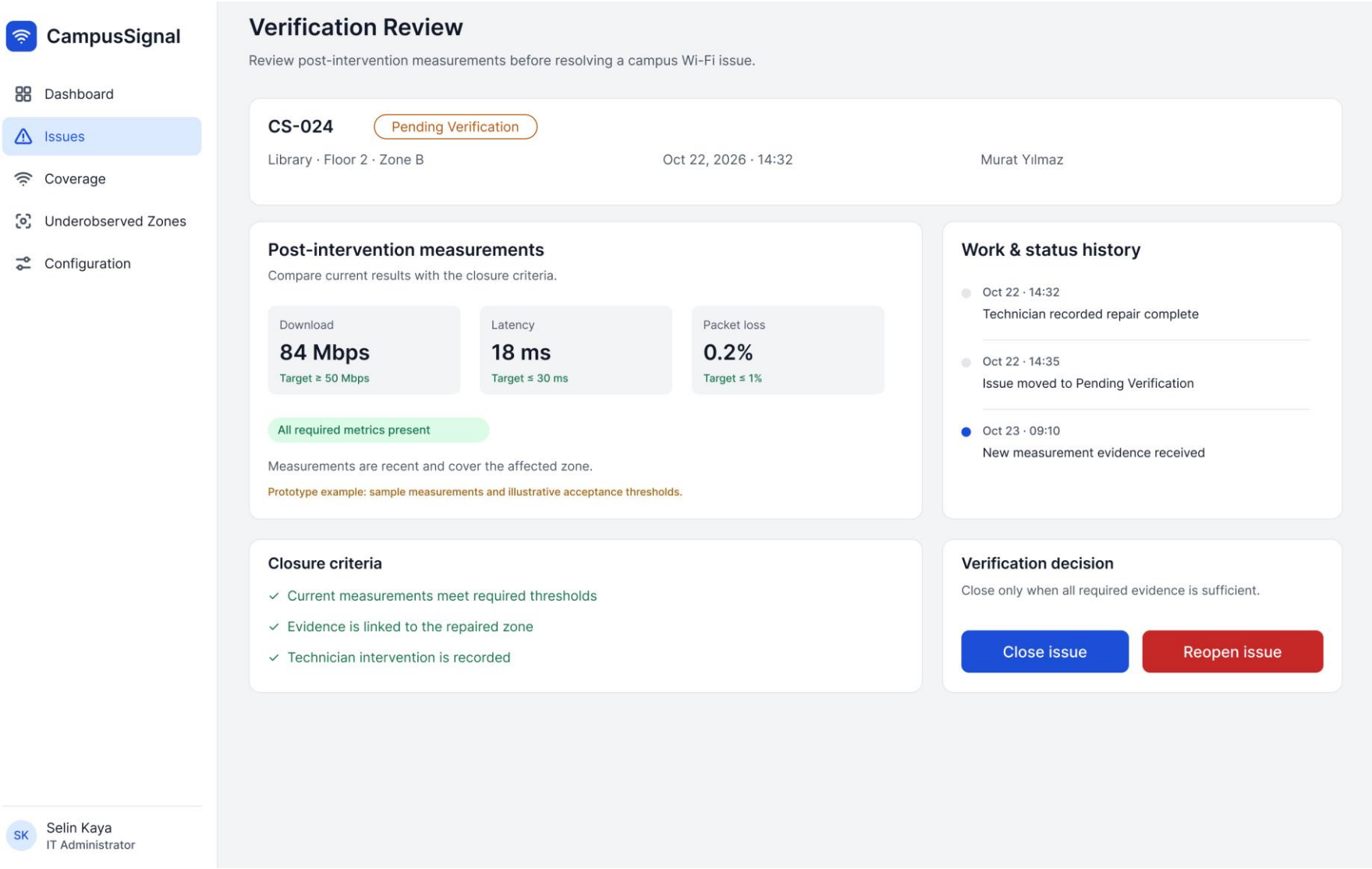


Figure 6. UC-04: UI-31 Verification Review. Closed, Reopened and Pending Verification remain distinct outcomes; a sufficient measurement count alone does not prove recovery.

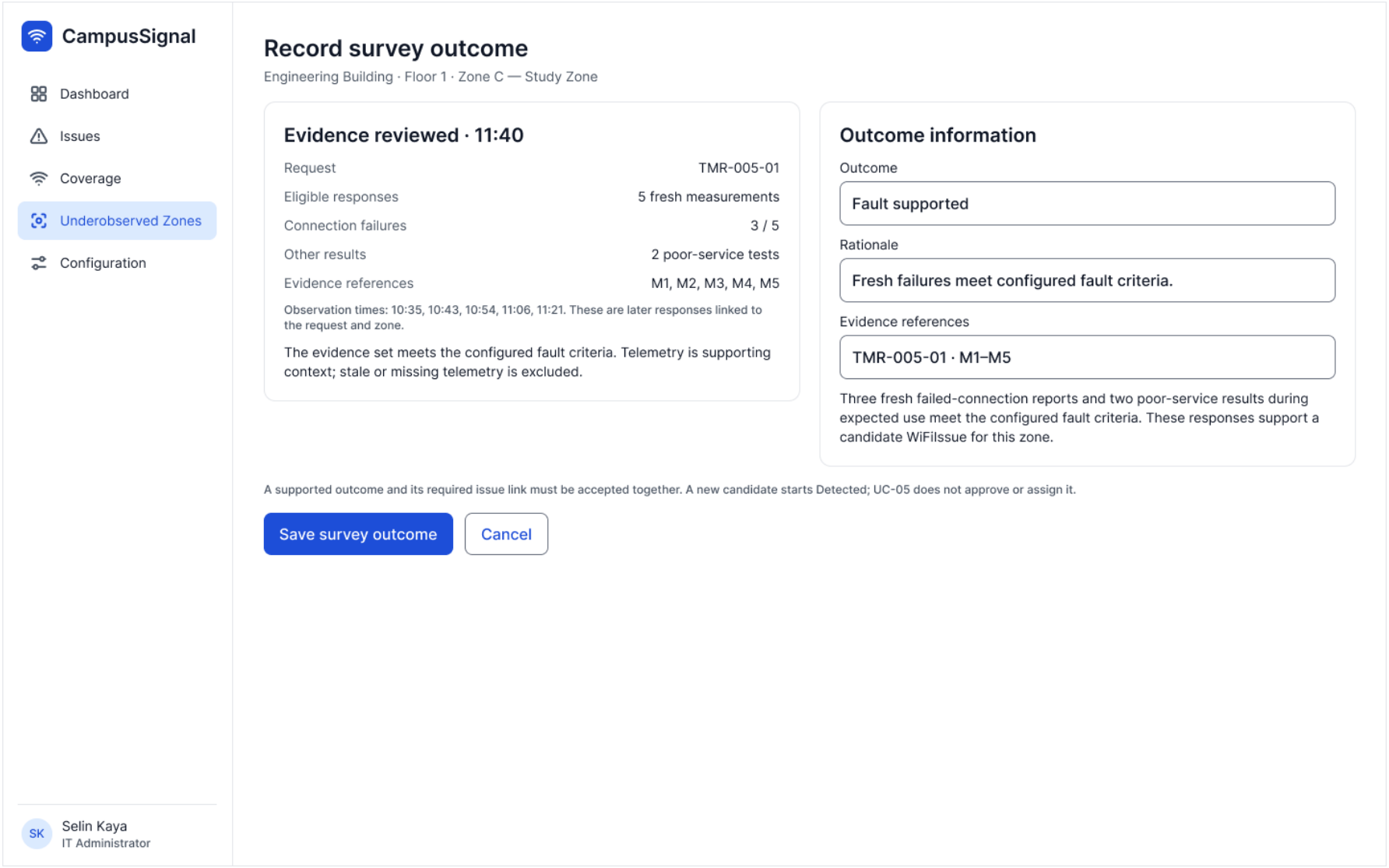


Figure 7. UC-05: UI-43 Record Survey Outcome. The Engineering / AP-ENG-03 example reviews five fresh responses before linking a new Detected candidate.

Prototype verification status

The 25 October screen export confirms the listed frame IDs and represented states. The UC-05 main path UI-40 → UI-41 → UI-42 → UI-42A → UI-43 → UI-43A was clicked in Present. Its 18 screens, 42 content buttons and 90 sidebar items passed the scoped link check. UI-43B and UI-43C represent the no-fault and insufficient-evidence alternatives.

UC-01 shows offline queue and synchronized-result snapshots. UC-02 includes rejection, Approved / Unassigned, assignment-save failure, notification failure and denied-access views. UC-03 includes diagnosis, intervention history, verification-request and recovery views; the Technician Coverage → My Issues link targets UI-54.

A dedicated Reopened → In Progress resume state is specified but has not been separately verified in the prototype. In UC-04, the current snapshots do not provide recorded click reactions for UI-30 Start verification test, UI-32 Back, UI-33 Reopen and UI-34 Return. These screens therefore illustrate the required states rather than certify a complete interactive recovery flow. No complete regression or backend acceptance test is claimed.

The submission package contains the final Word and PDF copies and the exported screen images. Efe's original UML diagram is retained in Figure 1. The original editable PlantUML source must be supplied by the diagram owner; a reconstructed reference is not a recovered original source.